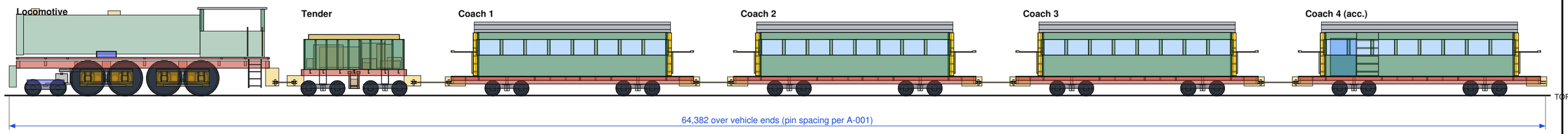
PULSIPHER FARMS CHRISTMAS TRAIN — DESIGN SITE DRAWING SET PF-CT REV L4.2

36 in gauge diesel-hydraulic locomotive + tender + 4 coaches · 6 mph GOVERNED (42 cc stroke stop + sealed 2,600 rpm; hardware 7.6 mph; 6.5 mph warning; trip 7 mph; 3 mph camera-fail) · 100 m min. radius · track LEVEL (Layout F 0.0 %), vehicles checked at 2 % (conservative)



GENERAL NOTES

1. PRELIMINARY. Owner engineering for review by a Nevada-registered PE; not for fabrication until stamped and released.
2. Codes (to verify by PE): Clark County Code Ch. 22.16 / Ord. 5378; ASTM F2291, F1193, F770, F3598 (amusement rides); AISC 360 (steel), AWS D1.1 (steel welding) / D1.3 (sheet), IBC 1607 for guard, rail and stair loads.
3. Materials: HSS ASTM A500 Gr C (Fy 50 ksi); plate & bar ASTM A36 unless noted; drawgear plate A572 Gr 50; pins and axles AISI 4140 Q&T 28-32 HRC (Fy >= 95 ksi, mill cert required). Mill certificates for all primary steel.
4. Welding: GMAW/FCAW E70 to AWS D1.1 by qualified welders under a written WPS. Default fillet = 6 mm all round unless sized on sheet. Primary drawgear welds (PF-DB boxes): 10 mm fillet all round, 100% visual + MT.
5. Bolts: structural ASTM F3125 A325 / SAE J429 Gr 8 / ISO 898 cl. 8.8-10.9 as noted; hardened washers; prevailing-torque nuts or Nord-Lock on running gear. Torque per fastener schedule, mark after torque.
6. Tolerances: ISO 2768-mK unless noted; cut lengths ±1.5, holes ±0.8, frame diagonals within 3 mm, pin bores H8. GD&T per ASME Y14.5-2018 where shown (L-103.2 final drive, axle seat); weld symbols per AWS A2.4 (schedule S-503).
7. Coupling height 550 mm ARL on all vehicles (empty, new wheels). Max wheel load basis 26 kN (K2-a 25.89 kN, L4.2). GOVERNED 6 mph max (owner-selected, PE to confirm).
8. Items marked AWAIT are waiting for supplier data (AIE/Kubota, Poclain, chain/sprocket maker, Danfoss/PLUS+1, IR, ZF/WABCO). Do not invent values.
9. Third-angle projection. Fastener torques per S-503 (0.75 x proof load; K 0.20 dry / 0.14 lubricated).
10. Calculations: PF-CT_PE_calc_package.pdf (rev L4.2) + calcs/L4/drive_L4.json. Open items / assumptions: PF-CT_assumptions_open_items.pdf.
11. Sheet numbers follow the Loco Bot shop-sheet master (L-1xx): L-103 final drive/motor mounting (+ L-103.2), L-105 hydraulic routing, L-106 front ballast; shop sheets L-103, L-105, L-106, L-107L, L-108, L-109, L-110, L-111, L-114, S-504, S-507, A-101 form part of this set (index). Track basis: Layout F LEVEL 0.0 %; vehicles checked at 2 %.

REVISION HISTORY

Rev	Date	Description
R1	2026-10-27	Initial issue for owner / PE review (12 sheets).
R2	2026-10-27	Owner decisions: emergency stairs all coach doors (C-304, F08);
		40 mph gust no-run + tie-downs DECIDED; roof 3,700 ADOPTED,
		cab headroom ~1,960, route overhead min. 4,300 ARL; pony open (rec. delete).
L1	2026-10-27	Owner: loco wheels 'like the plans'. Running gear per LT18-LT20:
		8x Ø1,450 drivers, 1,550 wb, inside frames, leading truck Ø650 (worn limit Ø630), unbraked, running
		board Z1,605. 2% start now motor-short at 1:1 (OPEN F25, C13).
L1b	2026-10-27	Owner: no chains (F25 chain options dropped). Model corrected to 4 motors,
		one per axle, staggered L/R per plan DT4 (was 8). Drive choice OPEN F25.
L2	2026-10-27	Owner chose F25 (c): inboard spur-gear case 21:38 (1.81:1) per axle; MSE05
		inboard, nose link to transom, pinion-shaft disc brake, NO sanders. Outboard
		DT5 cartridges, couplings, motors removed. F03, F25 RESOLVED (L-101, L-103).
L2	2026-10-27	PE-review pass: title blocks w/ tolerance + PE stamp box; added L-106, S-501..503,
		B-601, E-701; rail basis 26 kN; chevron springs; HSS3x2 roof bows; ERVs.
L2s	2026-10-27	Supplier fold-in: H1P053 G1 shaft + anti-stall, L35 350/400 bar, charge <=30 bar,
		case drains, spline motor shaft, eta 0.92/0.78, SKF gear-case brgs, Coremo ref.;
		ERV RenewAire HE05IN roof pod, 62.1 OA (E-701).
L3	2026-10-28	Owner: best fit. Catalog Dodge TA7315H15MS 14.9:1 + Danfoss H1B110 via 1:1
		belt drop (L2 custom case rejected); Ø120 seat 34CrNiMo6 ind.-hardened; axle
		disc + 2 calipers; TA rod to transom. GOVERNED 6 mph (owner, PE to confirm).
		L-101, L-103, L-106 (replaced), S-501, S-503, B-601, G-001.
L4	2026-10-28	Owner: no belt. Poclain MS05 468/234 2-speed + T04 brake, frame-mounted;
		ANSI 120-2 15T:45T 3:1 chain in oil bath; Ø150 pressed hub, no key; (flow divider
		dropped L4.1); stroke stop 6 mph; independent 7 mph trip; 650 kg front
		ballast + 1.25 t sprung pony; cyl. Y±790; IR 2340 + 14 kW cooler on tender.
		L-101, L-103, L-106 (replaced), T-201, K-401, S-501, S-503, B-601, G-001.
L4.1	2026-10-28	RELEASED. NO flow divider: PLUS+1 traction control (pony ref., destroke; sand deleted).
		Shaft code 5 W55x3x17; 8.8 fitted ring bolts 219 N.m; hub SHRINK 0.15-0.20; low traction
		pin h 0.65; trip ramp 3 s + 2nd default + T04 switch/choke; bump stops ±22; 38 cc stop;
		ballast 2 × 75.4 + clamp bars. L-101, L-103, L-105, L-106, S-501, S-503, B-601, G-001.
		s7: T04 release from loop (LP shuttle, 20 bar PRV, NV solenoid, choke + 1 L acc.; manual 6 L
		acc. + hand pump), no 4th dry-break; condemn drivers Ø1430, flange ≤ 30 mm; reverse ≤ 1.55 %.
		s8/9: tender TB-3.9 8,616.3 kg (0.676 m/s²); loco release ≤ 4 s (C ≥ 3.0, 3/4 in trunk, 1/2 in relay);
		safety-chain lugs PL40 CJP + PL16 diaphragms (+10.2 kg). B-601, S-503, S-501.
L4.2	2026-10-28	Stress Bot r3 rulings: FLOAT descents (bypass, >= 3 s ramp); T04 from charge, Ø1.2 orifices, 2-ch switch;
		42 cc + sealed 2,600 rpm; flange 28 / re-profile <= 27; traction pin Ø120 + GE120ES; sand system DELETED (dome cosmetic);
		reverse start <= 1.20 % (28.9 kN cap); HPRV 380, loop parts >= 420 WP, strength 430; chain cert >= 326 kN;
		4140 15T; NB-1 59.8 kg; lugs PL44.5; tender TB-3.13d; loop hose 560 bar; tow valve DN06 500; renumbered L-103.2, L-106.

SHEET INDEX

Sheet	Title
G-001	Cover, general notes and sheet index
A-001	Train general arrangement
L-101	Locomotive general arrangement
L-102	Locomotive frame and member schedule
L-103	Powered truck + final drive / motor mounting (Ø1,450, MS05 + 120-2 chain)
L-103.2	Final drive unit - MS05 + 120-2 15T:45T, 4140 hub, GD&T (was L-106)
L-104	PF-DB drop drawbar box, drawbar, pin, chains
L-106	Front ballast pockets (P-106), cab steps and grab irons (was L-105)
T-201	Tender frame and general arrangement
C-301	Coach general arrangement
C-302	Coach underframe and member schedule
C-303	Coach truck
C-304	Coach emergency / boarding stairs (R2)
K-401	Track, rail and clearance outline
S-501	Bill of materials - locomotive
S-502	Bill of materials - tender T2, coach R2, drawgear
S-503	Weld and fastener schedule
B-601	Brake, air and hydraulic schematic
E-701	Electrical one-line and coach HVAC
	LOCO BOT SHOP SHEETS (master numbering; part of this set):
L-103 shop	Final drive + motor mounting shop detail (P-103-1 hanger)
L-105 shop	Loco hydraulic routing (loop parts >= 420 bar WP)
L-106 shop	Front ballast (P-106-1 clamp bars, P-106-2 pockets)
L-107L	Running-board supports, light option (28 HB16)
L-108	Boiler barrel shell and rings
L-109	Smokebox, front and pilot (NB-1 nose ballast)
L-110	Domes, stack and fittings
L-111	Cab structure C1 + FT-1 tow point
L-114	Loco electrical/pneumatic interfaces
S-504	Loco body BOM + cut lists
S-507	Loco weld takeoff
A-101	Loco build and assembly sequence (weigh-in hold point)

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances

Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)

Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°

Machined: Ra 3.2 · bores H7/H8 as noted · break edges

GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Cover, general notes and sheet index

Sheet G-001 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27

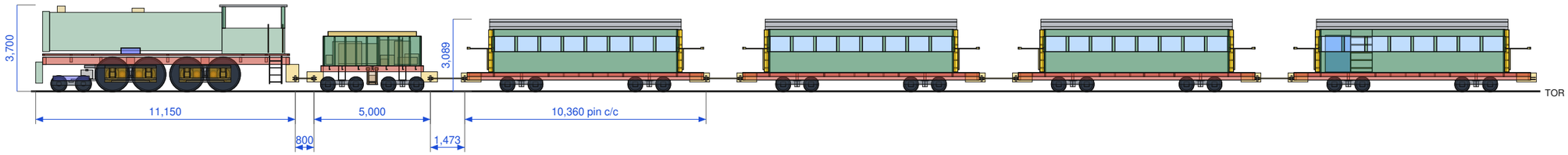
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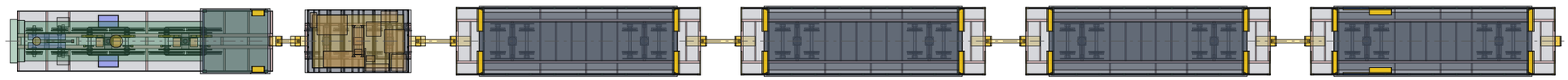
PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

SIDE ELEVATION



PLAN



CONSIST

Vehicle	Mass kg	Length	Width	Height ARL	Note
Locomotive	16,810	11,300	2,600	3,700	8 Ø1,450 drivers, 2 powered trucks + lead truck (LT18-20)
Tender	8,160 ledger / ≈8,300 rev T2	5,300	2,600	2,560	2× coach-type trucks; LT45.1 + TW-1 (T2)
Coach ×3 regular	13,200 loaded / 8,040 empty	10,160	2,794	3,089	32 seats
Coach ×1 accessible	11,000 loaded	10,160	2,794	3,089	11 seats + 2 wheelchair bays
Train	74,760 best est. / 77,500 design / 80,000 check	64,182			1.5 t ballast option incl. in design mass

KEY RULES

Coupling height 550 mm ARL, all vehicles.
Drawbars: DB-800 (loco-tender), DB-1473 (all others).
Two Grade 100 5/8 in safety chains + Crosby G-2130 1 in shackles at every joint.
Brake pipe + 8 bar feed pipe through every vehicle.
Min. radius 100 m (tangent at stations).
Clear 2,000 mm from track CL to fixed objects (K-401).
Overhead: nothing below 4,300 mm ARL within 2,000 of CL.
Egress strip 2,000-3,300 from CL walkable, >= 1 side.
NO RUNNING in gusts > 40 mph; tie down when parked (DECIDED).

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

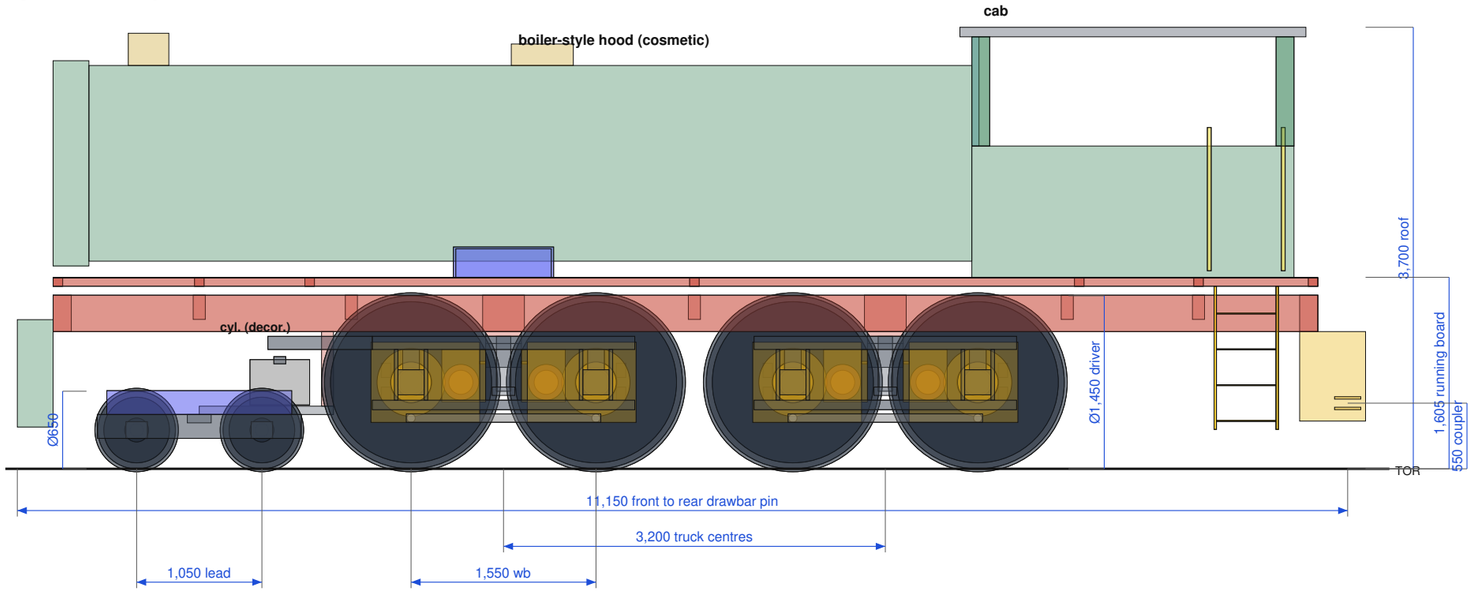
Train general arrangement

Sheet A-001 · REV L4 · 2026-10-28 · 1:200 at 17x11

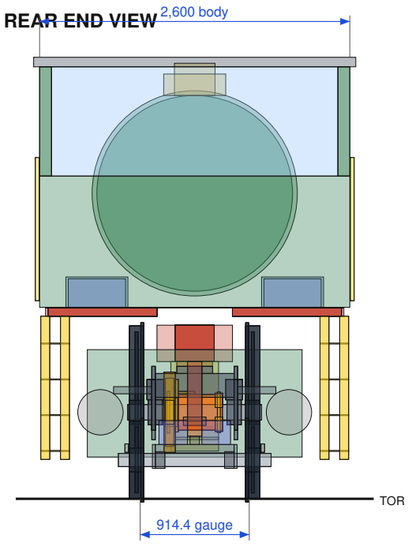
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

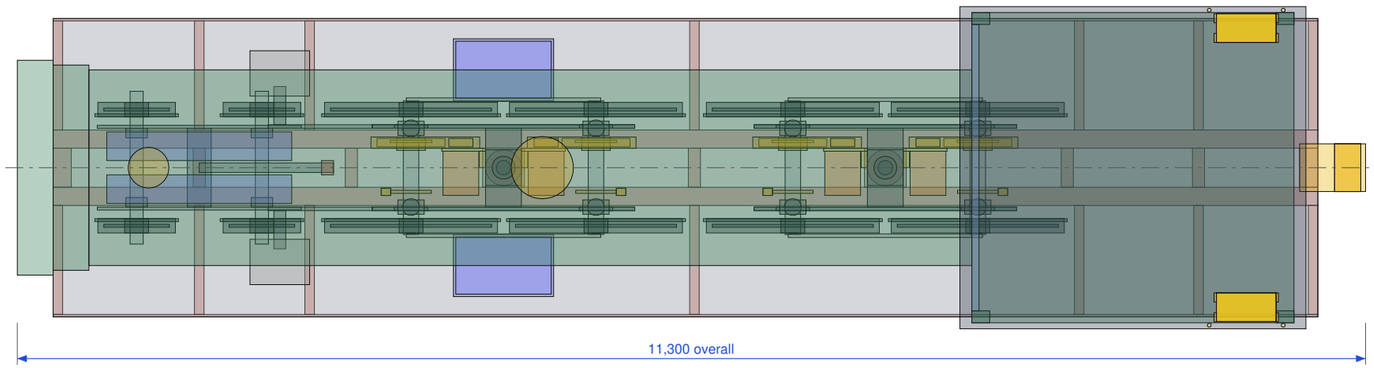
SIDE ELEVATION



REAR END VIEW



PLAN



NOTES

REV L1 - running gear per owner plan LT18-LT20 (to LT42):
8 drivers Ø1,450 (spoked look), 2 powered trucks, 1,550 wb,
pivots X4075/7275, axles X3300/4850/6500/8050; inside frames.
Leading truck Ø650, X1000/2050, swing arm to X2600. No trailer.
Main rails Y±240 Z1150-1455; running board Z1,605 (155 over drivers).
REV L4 drive: 4x Poclain MS05 468/234 2-speed + T04, frame-mounted,
ANSI 120-2 15T:45T 3:1 chain in oil bath, Ø150 pressed hub (L-103.2).
NOTHING outside the wheels; frame between the wheels.
2% start 37.67 kN @350 bar vs 32.61 (NO divider; equal torque dry 1.14,
1.146 with truck-pitch transfer, low traction pin h 0.65). Sand DELETED: wet-start O&M rules.
GOVERNED 6 mph (42 cc stop + sealed 2,600 rpm: 5.96 mph); indep. 7 mph trip.
Brakes: axle disc + 2 spring calipers per powered axle (Y+); MS05 T04 park.
Ballast: 151 kg front pockets + NB-1 59.8 kg nose (X450); pony 1.25 t.
Max driver wheel 25.89 kN <= 26. Decor. cylinders Y±790.
Fixed cab steps (L-106).
Roof 3,700 ADOPTED. Cab floor 1,605, headroom ~2,010 w/ mat.

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Locomotive general arrangement

Sheet L-101 · REV L4.2 · 2026-10-28 · 1:50 at 17x11

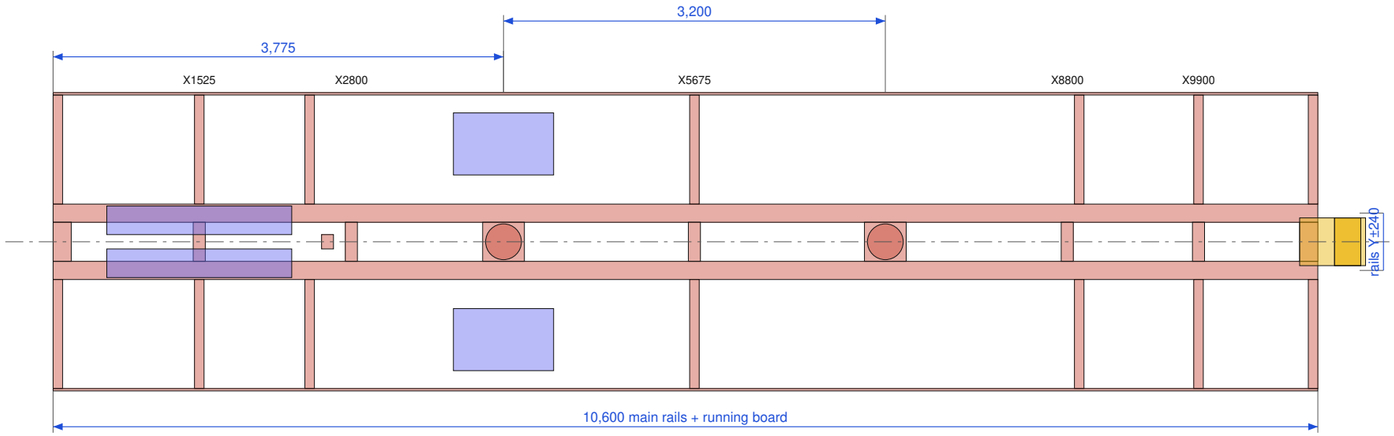
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

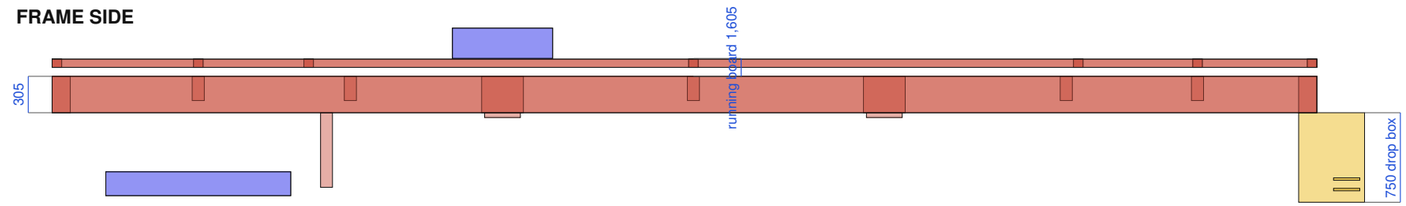
PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

FRAME PLAN (deck plate removed)



FRAME SIDE



MEMBER SCHEDULE

#	Part	Section / material	Qty	Length	Location
1	Main rail	HSS12x6x3/8 A500C	2	10,600	Y±240, Z1150-1455 (LT18)
2	End sill	HSS12x6x3/8 A500C	2	328	X300 / X10,748
3	Crossmember	HSS8x4x1/4 A500C	5	328	X1525/2800/5675/8800/9900
4	Body bolster box	PL12 flg 350 + PL10 web	2	328	X4075 / X7275
4a	Running-board outrigger	C3x4.1 A36	14	914	X340/1525/2450/5675/8900/9900/10860
4b	Board edge angle	L3x3x1/4 A36	2	10,600	Y±1,240
4c	Swing-arm anchor	PL20 box	1	625	X2600, leading truck
5	Centre plate	PL40 Ø300 + Ø60 pin bore	2	-	under bolster
6	Drop drawbar box	PL16 A572-50 box 552x400	1	750 tall	pin X11,150 Z550
7	Knee gussets	PL16 A572-50	4	600	box to each rail
8	Running board / deck	PL6 tread plate	1	10,600x2,500	top Z1,605
9	Ballast pocket	PL20 ends / PL12 sides	2	840x520x255	over front bolster

NOTES

Hand checks (calc notes C4, C4b, C5, C8):
rails 90 MPa at 2 g; 225 MPa at 200 kN drawbar (< Fy 345).
bolster 34 MPa at 2 g; drop box 34 MPa / weld 85 MPa at 200 kN.
All crossmembers cope-fit, 6 mm fillet all round.
Rails: no splices. Camber none. Check diagonals within 3 mm.
Lifting: 4 jacking pads under bolster ends (load 2 g).
Equipment hard-points to be added from AIE/Poclain data (AWAIT).

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Locomotive frame and member schedule

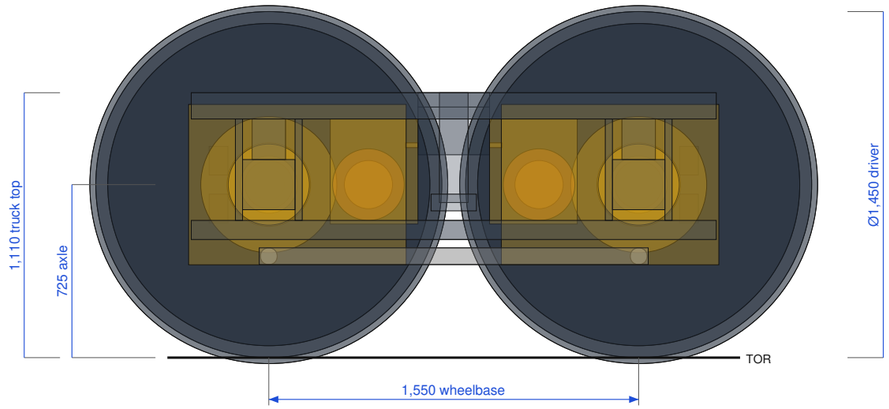
Sheet L-102 · REV L2 · 2026-10-27 · 1:50 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

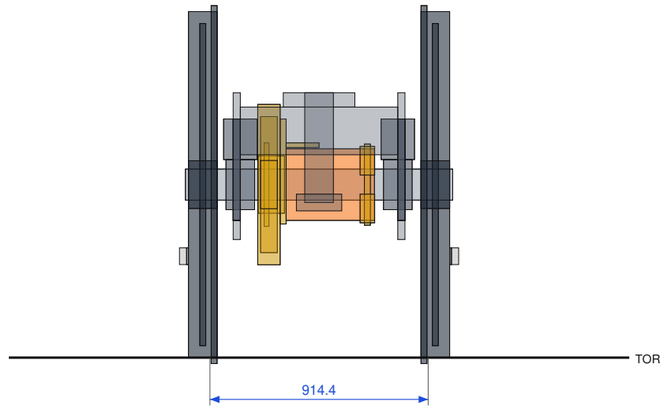
Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

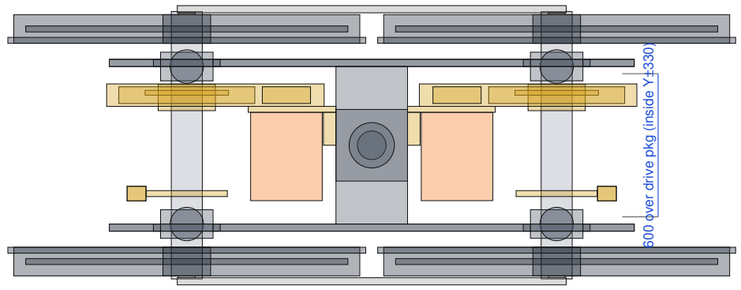
SIDE



END (section at axle)



PLAN



NOTES / FIXES

REV L1: plan LT18-LT20 truck - inside sideframes Y±330-360,
top Z1110; drivers Ø1,450 exposed outside; flange gap 50 in truck.
Transom PL16 box; pivot seat + Ø100 spherical pivot (LT20 GE100ES-2RS cand.).
REV L4 per axle (Y-): 45T 120-2 split rim ring bolted to a Ø150 pressed flanged hub
(no key, SHRINK 0.15-0.20), 54-link chain C417.6 to a 15T
on MS05 shaft code 5 W55x3x17; motor on PL25 slotted bracket under transom
(4x M20 10.9, 2 jack screws, 37 mm slide); oil-bath case Y-257..-163,
Z389-1061 (>=15 radial at 3% + travel), floating seal plate. Clear: case 3 / hub 5 to
spring reservation, motor 57 (23 at 3% stretch) to transom.
Disc Ø340x25 on axle Y+ + 2 spring calipers (B1-g), unchanged.
Removed: Dodge TA7315H15MS, H1B110, belt drop, TA rod.
AWAIT: Poclain AE (radial/offset, spline); chain/sprocket maker.
Inside axleboxes AWAIT bearing; primary springs AWAIT (LT35).
Anti-slip (L4.1): NO flow divider - PLUS+1 traction control, pony 3rd pickup (sand deleted).
L4.2: LOW TRACTION PIN Ø120 4140 + GE120ES housing under transom at Z650
(X4075/7275, calc L42-tp); centre plate flat-lined (vertical only). Bump stops ±22.
Sanders DELETED (11/2026); hill-hold kept; flange withdraw 28,
re-profile <= 27 (all wheels); P-103-1 hanger L-weld + 2 PL16 gussets (Loco Bot L-103).

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

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Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Powered truck + final drive / motor mounting (Ø1,450, MS05 + chain; detail L-103.2)

Sheet L-103 · REV L4.2 · 2026-10-28 · 1:25 at 17x11

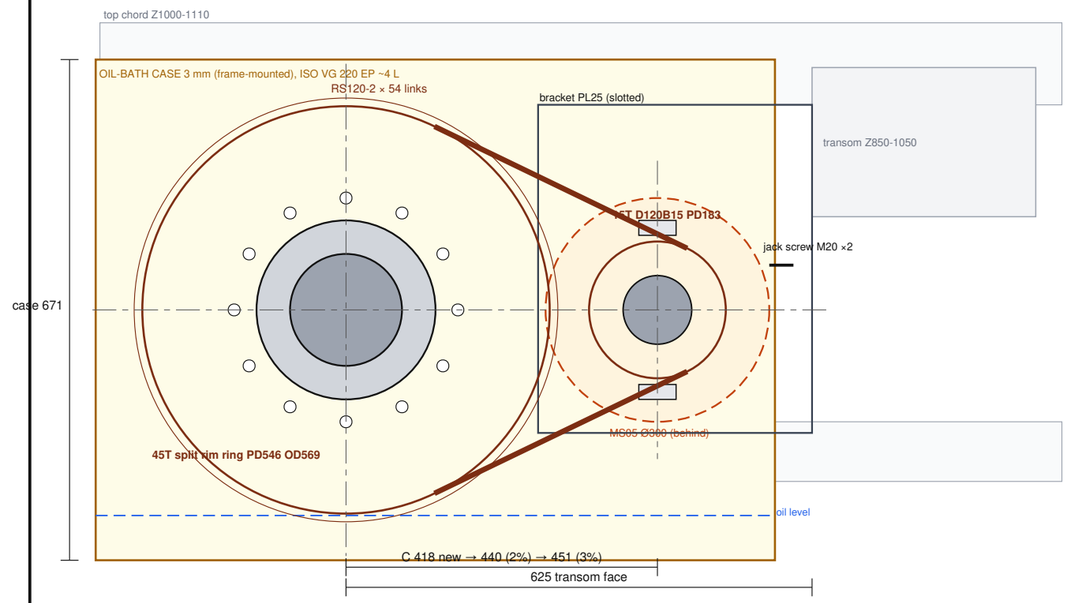
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

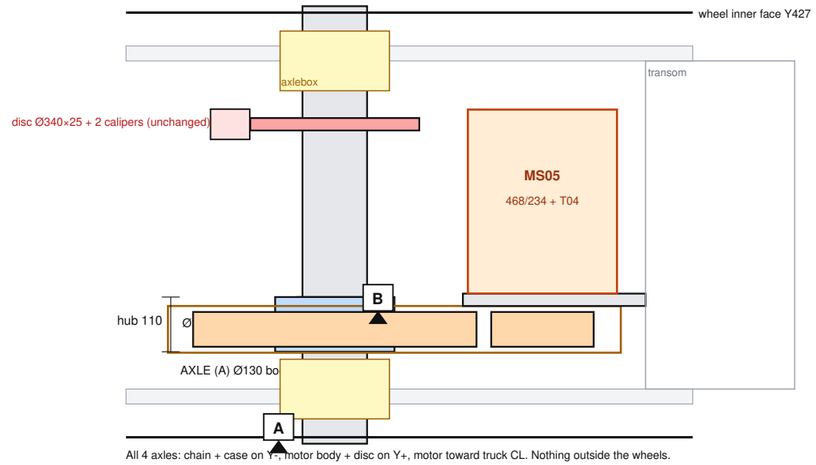
REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

VIEW A - SIDE (from Y-, chain case cover removed; truck centre to the right)



Motor at 3% stretch clears the transom by 23 mm; case bottom Z389 ARL; chain line horizontal (±25 mm travel → +0.75 mm centres): set sag 17-20 mm, no idler.

SECTION B-B 1:12 (plan through axle CL; chain/motor flange Y-, motor body + disc Y+)



All 4 axles: chain + case on Y+, motor body + disc on Y+, motor toward truck CL. Nothing outside the wheels.
Seal plate: lip seal on hub land, PTFE-faced plate slides ±25 mm vertical / ±3 lateral in a case guide (OPEN OI-L4-5).

BILL OF MATERIALS - ONE DRIVE UNIT (×4 per loco)

Item	Description / material	Qty	Source
1	Poclain MS05 2-disp. 468/234 cc HighFlow, T04 brake, shaft code 5 DIN 5480 W55x3x17, speed sensor 56 ppr	1	AWAIT Poclain
2	Sprocket 15T 120-2 OD 202, MADE-TO-ORDER 4140 Q&T hub (L4.2), DIN 5480 N55x3x17, ≥ 58 engaged	1	AWAIT maker
3	Sprocket 45T 120-2 rim-only SPLIT ring, 2 halves (~50 kg, ASSUMED), 12 × M16 8.8 fitted, 219 N·m, to Ø350 hub flange; replace in place	1	AWAIT supplier
4	Hub 4140 Q&T Ø240 × 110 + Ø350 × 20 flange (31.4 kg), bore Ø150, SHRINK ONLY 0.15-0.2 at 200-230 °C; removal oil groove	1	machine shop
5	Chain ANSI 120-2, mill cert. MBL ≥ 326 kN per lot (L4.2; std RS120-2 296 kN NOT accepted), 54 links + conn. link	1	AWAIT chain maker
6	Oil-bath case 3 mm + floating seal plate + lip seal, sight glass, breather, mag. drain	1	fab / AWAIT seal
7	Motor bracket PL25 A572-50, slotted (4 × M20 10.9 in 50 slots), 2 jack screws M20, gussets	1	fab
8	Axle 34CrNiMo6 Q&T, Ø150 raised seat (no keyway, no hardening credit), UT + MT	1	machine shop
9	Disc Ø340×25 + 2 spring-applied air-released calipers (unchanged)	1+2	AWAIT brake sup.

GD&T (ASME Y14.5-2018)

CYL	0.008	-	Axle sprocket seat Ø150 (DATUM B) - hub bore for 0.15-0.2 mm diametral interference, Ra 0.8
↗	0.020	A	Seat radial runout to bearing journals (DATUM A)
↗	0.03	A	Seat shoulders r13 + undercut at hub edge; NO keyway; oil-injection groove + M8 port
↗	0.25	B	45T plate pitch circle runout to hub (fitted bolts PCD 300)
▱	0.40	B	15T and 45T tooth faces coplanar (chain alignment)
⌷	0.30	A	Motor slide plate square to axle in plan (slots parallel to chain line)
↗	0.05	A	Disc seat - runout

NOTES

1. 2% start: need 32.61 kN; 37.67 kN @350 bar (η 0.90). NO divider: PLUS+1 traction control; equal torque 1.14 dry,
1.146 with truck pitch (traction pin h 0.65; F2 37.11 kN, ballast 2 × 75.4 kg). Sand DELETED (O&M wet rules).
2. L4.2: chain pull 35.2 kN at the 430 bar strength basis (HPRV 380) → certified MBL ≥ 326 kN (factor 9.24); 296 kN std chain NOT accepted.
3. Seat Ø150, NO hardening credit, fretting Kf 2.0, heaviest axle 49.12 kN: Goodman n 1.57 ≥ 1.5. Fit slip 29.9 kN·m = 3.11× HPRV.
4. GOVERNED 6 mph: stroke stop 42 cc sealed + HARD tamper-sealed 2,600 rpm → 5.96 mph in 2nd; indep. 7 mph trip; hardware max 7.64 mph.
5. Single-axle spin 421 rpm at high idle (2nd) vs HighFlow 430 → HighFlow REQUIRED + traction control.
6. Motor radial 35.2 kN vs ~60 kN (code 5 chart) - sprocket close to flange; AWAIT Poclain AE. 2A40 keyed REJECTED.
7. Upkeep: oil weekly, change 500 h/season; sag monthly (17-20 mm at ride height); stretch 250 h; REPLACE CHAIN AT 2% (hard). Re-shim at centre plate only; primary ≤ 3 mm + slide reset. Unsprung drive 127 kg/axle.
8. T04 = parking only; L4.2 release from the H1 CHARGE circuit (M3) via 20 bar reducer (loop shuttle deleted); Ø1.2 orifice per port, hard/4SH lines; 2-ch switch vents BP first.
9. Chain: cert. MBL ≥ 326 kN per lot; bump stops ±22; reversal only at zero speed + 1 s ramp. Descents in FLOAT (loop bypass, ≥ 3 s ramp).
10. Wheels: condemn drivers at Ø1430; flange height WITHDRAW at 28 mm, re-profile ≤ 27 (train-wide, L4.2); enter diameters in PLUS+1.

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Final drive unit (L-103 sh 2) - Poclain MS05 2-speed + ANSI 120-2 15T:45T, 4140 hub (GD&T)

Sheet L-103.2 · REV L4.2 · 2026-10-28 · 1:8 at 17x11

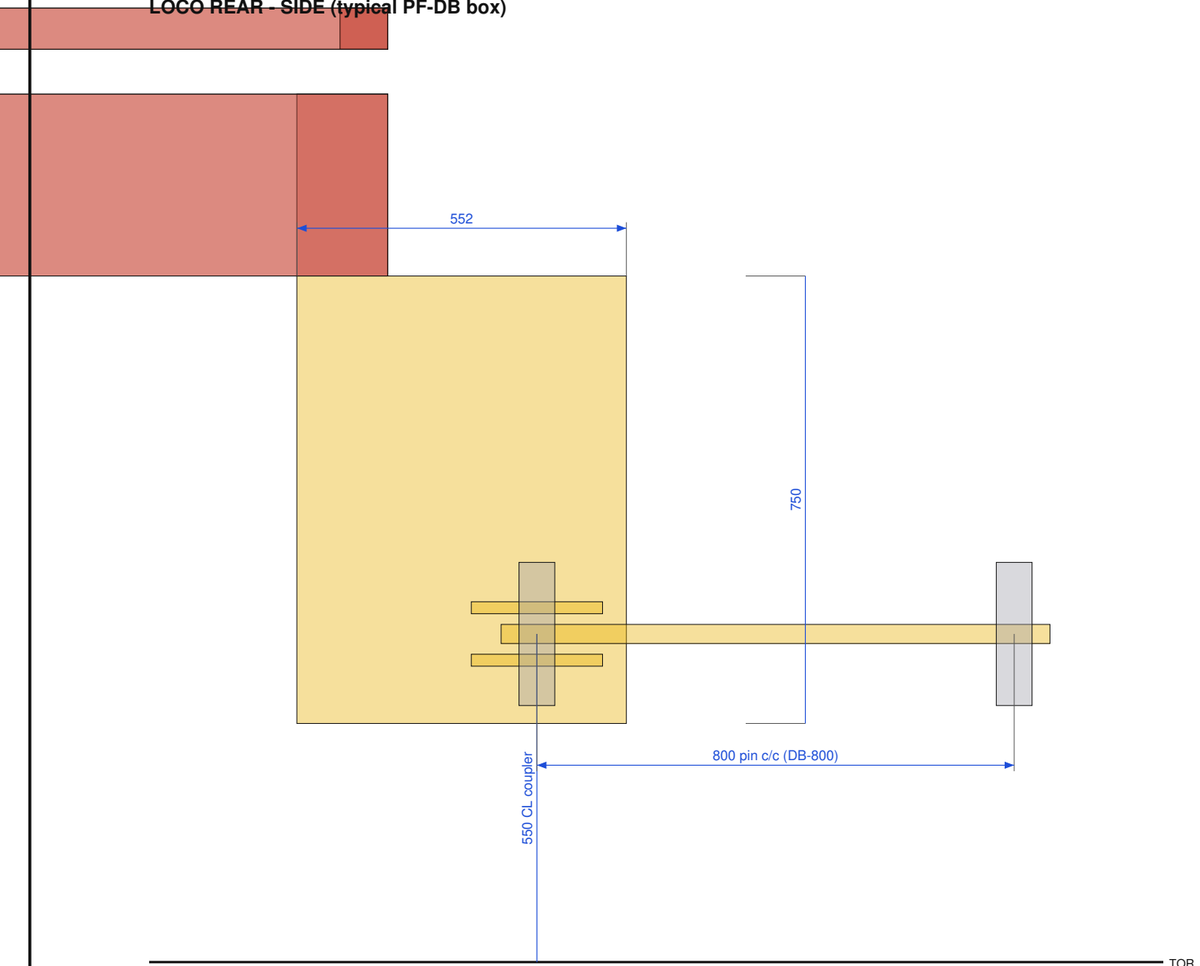
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

LOCO REAR - SIDE (typical PF-DB box)



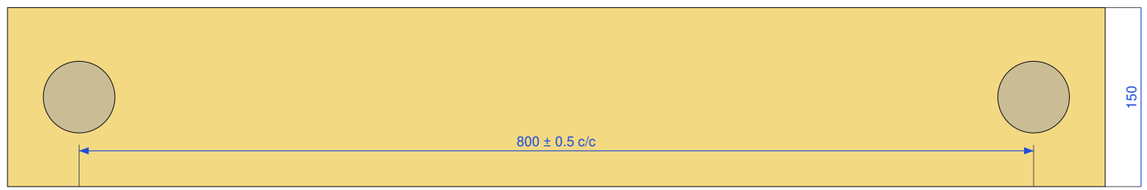
DRAWGEAR SCHEDULE (one design, used on every vehicle)

Item	Spec	Size	Note
DB-800	PL32x150 A572-50	800 c/c	Loco-tender
DB-1473	PL32x150 A572-50	1,473 c/c	Tender-coach, coach-coach
Pin	Ø60 4140 Q&T, H8/f7	L=150 u/head	grip 108 + head 20 + Ø10 cross-bolt + R-clip
Bushing	Ø61/Ø75 spacer, 4140	2 per pin	limits gap to 4 mm
Chains	2× G100 5/8 in, WLL 22,600 lbf ea.	slack 150	2× G-2130 1-1/8 + 2× S-1325A (L4.2); loco lugs PL44.5
Box	PL16 A572-50	400 wide	10 mm fillet all round + MT

WHY / CHECKS

Loads (owner basis, PE to confirm): ±35 kN normal; 70 kN no permanent deformation; 200 kN ultimate (no rupture).
Checks: drawbar net 70 MPa, bearing 104 MPa, buckling 2.7×; pin bending 104 MPa, shear 35 MPa; plates bearing 83 MPa; box 34 MPa, weld 85 MPa (all at 200 kN). Chains MBS 804 kN.
Why: replaces the 752 mm eccentric rear mount and the 400 mm tender offset with one closed box that carries the moment straight into both rails via knee gussets.
Truck restraint (I-08) stays with the PE: centre-pin retaining collar + 2 safety straps per truck shown on L-103/C-303.

DB-800 PLAN 1:5



PE STAMP / SIGNATURE

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Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

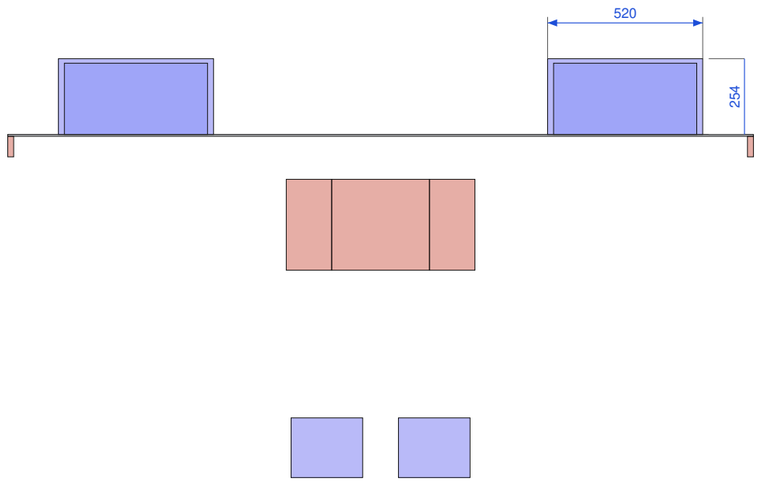
PF-DB drop drawbar box, drawbar, pin and safety chains

Sheet L-104 · REV L4.2 · 2026-10-28 · 1:10 at 17x11

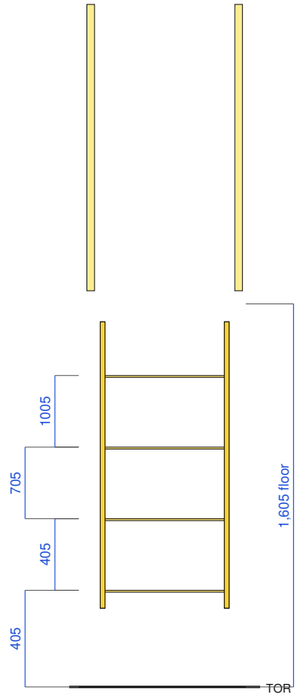
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

BALLAST POCKETS - SECTION AT FRONT BOLSTER



CAB STEPS + GRAB IRONS - SIDE



BALLAST (bolt-on)

Each side: pocket per Loco Bot L-106 (P-106-2: 60 walls), welded to deck/rail.
Fill (L4.1): 1 plate PL25.4 800x473 A36 (75.4 kg) per side.
Hold-down (P-106-1, L4.2): 4 × FB100x16x475 clamp bars 106-CB1 (23.9 kg)
inside the pocket on M16 set screws 4.8 N·m + jam nut 60 N·m; M20 rods 120 N·m.
BT-1 68.2 kg in Y- pocket. Check C9: end wall at 3 g (Loco Bot L-106).
NB-1 59.8 kg nose ballast PL25.4 at X450 (detail Loco Bot L-109).
Effect: K2-a 25.89 kN (fwd TE limit 33.0 kN) / 25.74 (rev 28.9) ≤ 26 kN.
Sanders DELETED - sand dome is cosmetic trim only. Weigh-in hold point A-101.

CAB STEPS

Replaces stowed ladder (I-09): nothing moves, door stays clear.
4 treads PL8 serrated, 500 x 240, risers 300, inside 1,300 half-width.
Stringers FB 20x75, hung from PL12 brackets on main rail (2x 1/2-13 Gr 8).
Grab irons Ø32 (1-1/4 in) solid A36, 1,200 long, both sides of
each door, 75 mm clear to body; checks 83 MPa at 200 lb.
Tread: 65 MPa / 1.7 mm at 300 lb point (C10a/b).
Bottom tread 405 ARL (rev L1, floor 1,605): station step-stool or platform.

PE STAMP / SIGNATURE

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Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

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Front ballast pockets (P-106), cab fixed steps and grab irons (was L-105)

Sheet L-106 · REV L4.2 · 2026-10-28 · 1:20 / 1:25 at 17x11

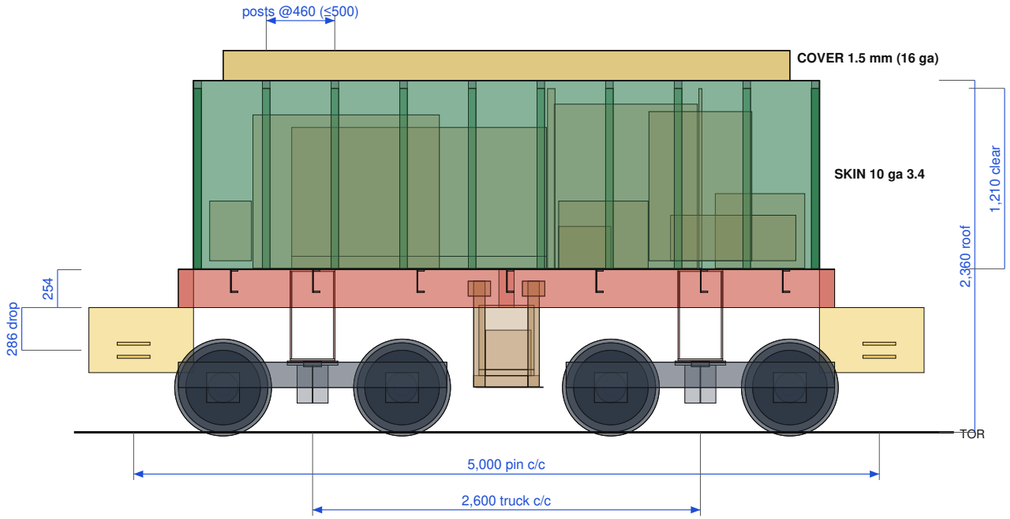
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

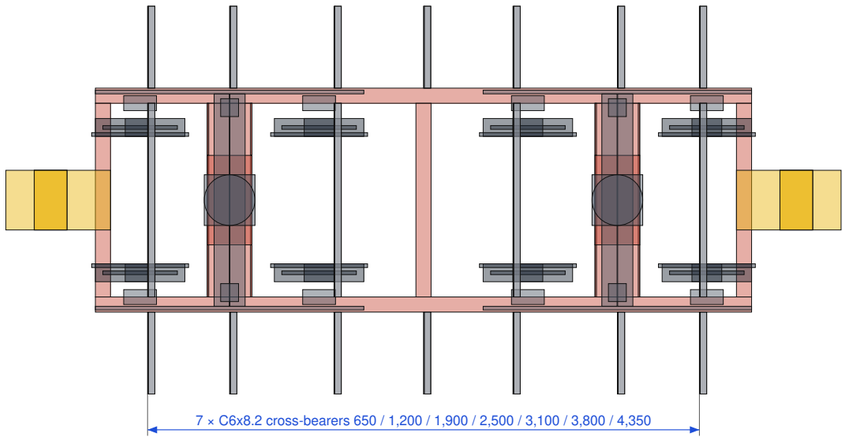
PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

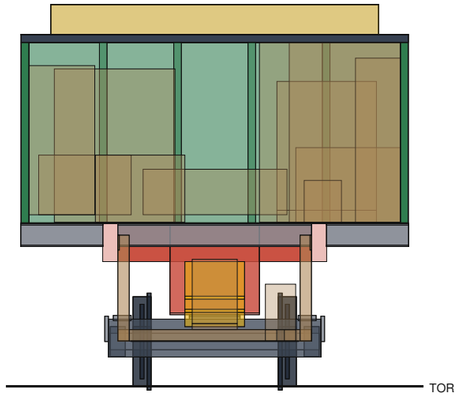
SIDE



FRAME + DECK PLAN (6 mm tread plate removed)



END



NOTES

Basis: owner plan LT45.1 (25 Oct 2026), two-engine tender. Rev T2 = TW-1 S1+D1+K1+B1.
Body X400-4600 x Y+-1300; deck Z1096 (rail top +6); roof Z2360.
Rails HSS10x4x1/4 Y+-700; fish-belly bolsters to centre plate.
PF-DB boxes both ends, 286 drop. Trucks = coach truck C-303.
LEFT: AIE Kubota V3307 + radiator + H1P053 (1,028 tall); cooler >=14 kW X3455-4145.
RIGHT (rev TB-3 approved): genset reservation X1060-2768 (≤ 1,775)
x 800 x 1,000 - PTI-38 699 kg at Y417-1084. Rated 40 °C not 43.
R wall NEMA 3R elec. cabinet X2820-3780, outward doors; listed UL 1008
ATS w/ integral main. Main bkr QOM22225NRB encl., handle <= Z1639. Batt X2850-3200.
Fuel 150 L custom Al X3900-4500 behind steel firewall X3800.
MR 3x Haldex 19820 X3600-4440; IR 2340 compressor L fwd X460-840.
Tender 8,993.3 kg (TB-3.15e, data/mass_ledger.json); CG [2433,+10,1109] on real T2 CG;
trucks 43.7/39.4 kN. LORD CB-1180-2 x4. XFMR assy 168.7 kg CG [2502,24,608]:
Acme T2535173S Z420-851 on isolators; L3x3 cradle, 2xM16 8.8 thru-bolts
/hanger at Z963 w/ crush tubes (no rail welds); shield 3 mm Z300 Y+-560.
Inrush box on cradle X2360-2665. Nothing under deck aft of X2900.
Unit top CLASHES X2500 xmbr 15 (K2-xf).

REV T2 SCHEDULE (owner approved 27 Oct 2026)

SKIN: 10 ga (3.4 mm) steel sides, ends, roof; bolted (round-head) or
plug-welded to HSS2x2x1/8 posts + roof bows at <=500 (riveted look).
No continuous seam welds on the sides (oil-canning). 50 mm lining kept.
DECK: 6 mm tread plate, sealed spill floor, on rails + 7 C6x8.2
cross-bearers (outboard to skin; between rails where no crossmember
or bolster). Equipment mounts land on rails / bearers.
COVER: coal-effect, 1.5 mm (16 ga) formed steel, removable, black
textured coal finish, gold edge trim. Height 200 ASSUMED.
BOLSTERS: fish-belly PL10 webs, PL12 flanges; full depth over Y+-300.
WHEELS: Ø600 plate-web, rim 50 / web 25 / hub Ø200 (section ASSUMED).
Tender mass SUPERSEDED by TB-3.15e 8,993.3 kg (see data/mass_ledger.json; xfmr assy 168.7 kg).

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Tender frame and general arrangement (rev T2 / TB-3 layout)

Sheet T-201 · REV L4 · 2026-10-28 · 1:40 at 17x11

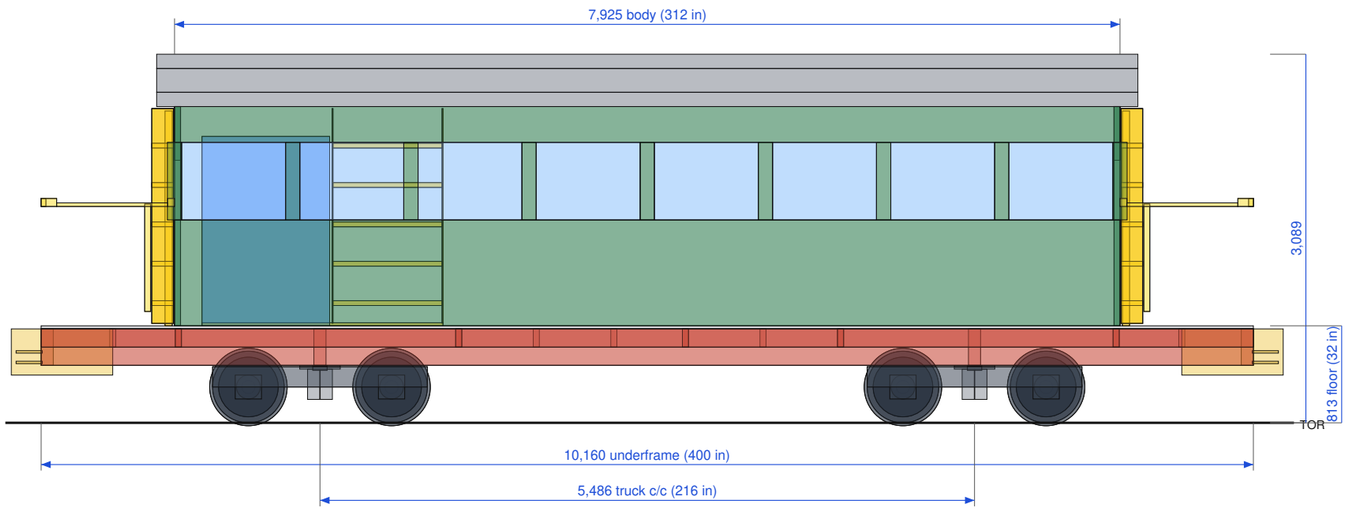
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

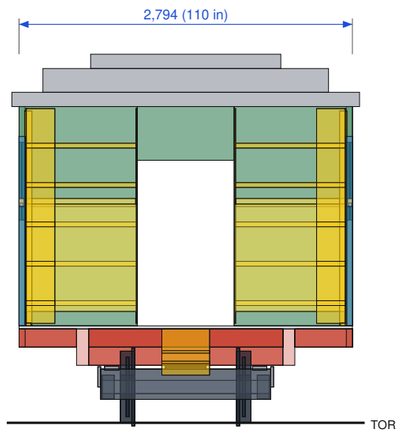
PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

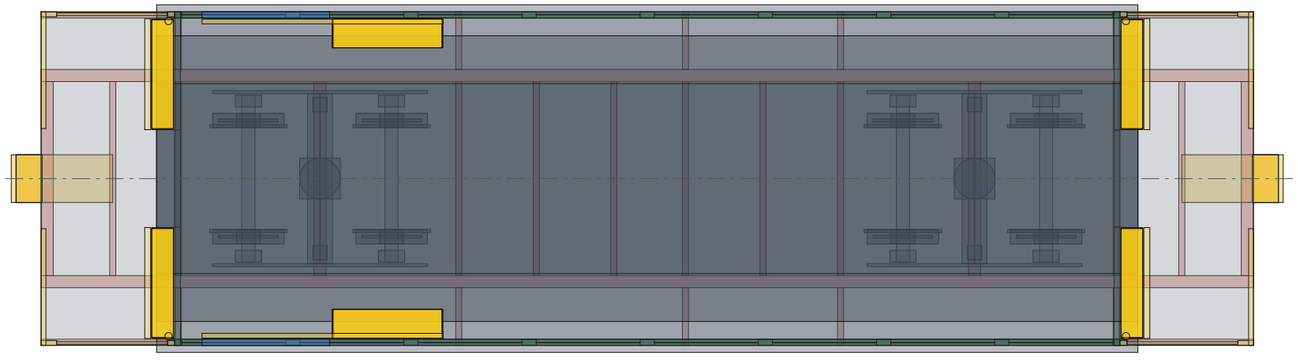
SIDE ELEVATION



END



PLAN (roof removed not shown)



NOTES

Regular: 8 rows 2+2, 32 seats, 24 in aisle (PE/County to confirm).
Accessible: 42 in side doors both sides, 11 seats + 2 bays.
Floor 32 in (813) ARL. Level 32 in platforms OPTIONAL (F08).
Emergency/boarding stairs at EVERY door, both sides (C-304):
S1 at end-platform side openings, S2 at 42 in side doors;
6 x 177 risers to ground, stowed inside body (old steps deleted).
End platforms 44 in with 42 in guards, 2 in sch 40 posts (C10d).
Wind DECIDED: no running in gusts > 40 mph (FS 3.85 empty);
tie-downs when parked outdoors (C11, C11w).
HVAC: mobile-rated roof/underfloor unit AWAIT selection (I-02).
Seats/upholstery: by upholstery trade on C21 bench frames.

PE STAMP / SIGNATURE

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Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

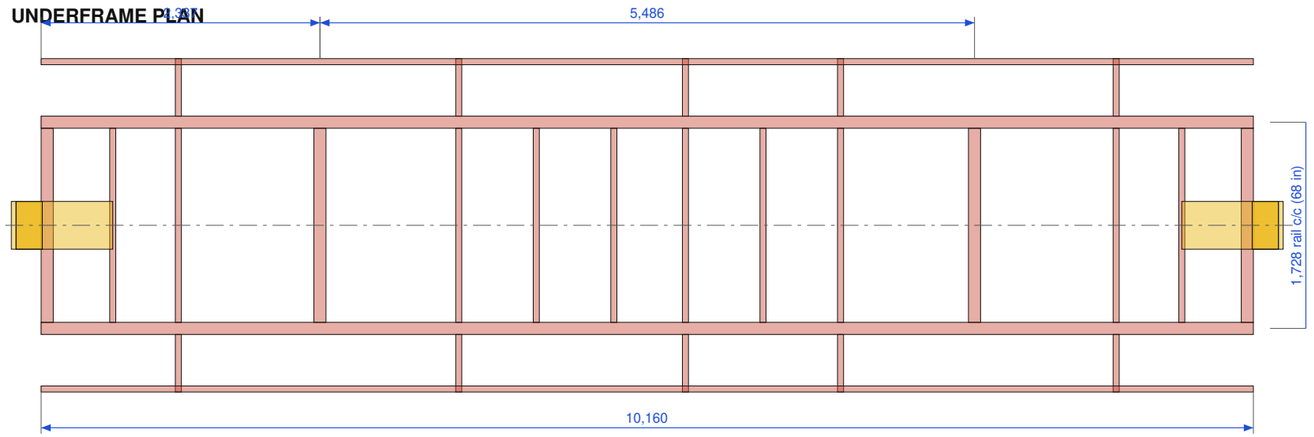
Coach general arrangement (regular; accessible variant noted)

Sheet C-301 · REV L2 · 2026-10-27 · 1:50 at 17x11

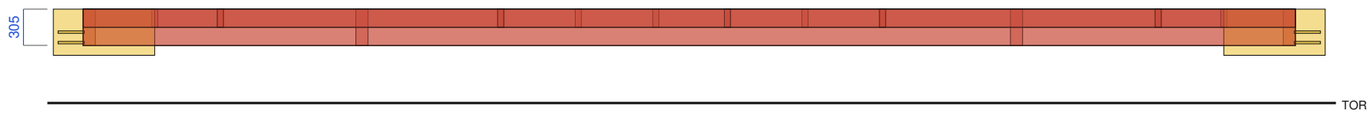
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

UNDERFRAME PLAN



SIDE



MEMBER SCHEDULE (matches J1a study sections)

#	Part	Section / material	Qty	Length	Location
1	Main rail	HSS12x4x1/4 A500C	2	10,160	Y±864
2	End sill	HSS12x4x1/4 A500C	2	1,626	
3	Body bolster	HSS12x4x1/4 + 2 PL10x100 + diaphragms	2	1,626	X2337 / X7823
4	Crossmember	HSS6x2x3/16 A500C	10	1,626	see plan
5	Outrigger	HSS6x2x3/16 A500C	10	432	to side sill
6	Side sill	HSS6x2x3/16 A500C	2	10,160	Y±1,371
7	PF-DB box	PL16 A572-50	2	850	pins X-100 / X10,260
8	Floor	1 in composite panel, screwed + bonded	1		

NOTES

Checks: main rail 48 MPa (×1.3), 74 MPa (×2.0); +70 kN buff 64 MPa.
Bolster 77 / 118 MPa; end deflection 1.7 mm.
Crossmembers kept clear of truck sweep (±1,000 of bolster).
Coupler CL 550 lies inside rail depth 483-787 → no drop needed beyond the 850 box; no eccentric hitch (fix F02).
All tube ends capped; drain holes Ø10 at low points.
Jacking pads under bolster ends (4).

PE STAMP / SIGNATURE

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Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
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GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Coach underframe and member schedule

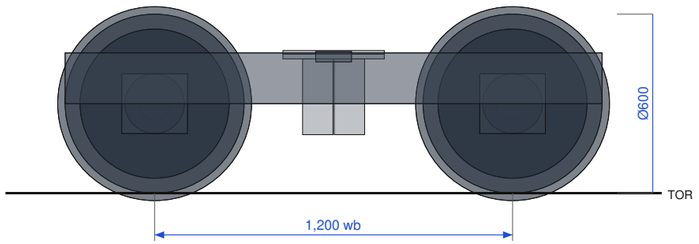
Sheet C-302 · REV L2 · 2026-10-27 · 1:50 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

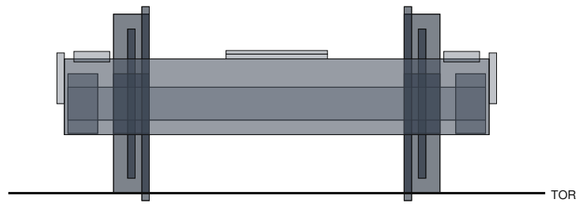
Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

SIDE



END



PLAN

NOTES

Sideframes PL25x170 A36; transom 2× HSS10x4x1/4 upright + PL16 cap.
Centre plate Ø340 PL12 with Ø60 centre pin + retaining collar.
Side bearers: replaceable UHMW/bronze pads, 5 mm clearance.
Axleboxes: SKF 22218 E (90x160x40) with KM18/MB18 - retained CW3.
Springs: 8 coils per truck, C5 rates (148 N/mm) AWAIT supplier.
Brake: one air-disc caliper per axle, PAN17-family 325 rotor,
spring-brake chamber; AWAIT ZF/WABCO data.
Hoses: relay valve on body; only 2 moving hoses per truck (fix F10).
Checks: transom 66 MPa, sideframe 80/123 MPa, axle 52 MPa.
Restraint: 2 safety straps truck-to-body (I-08, PE).

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

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GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Coach / tender truck (plate fabricated)

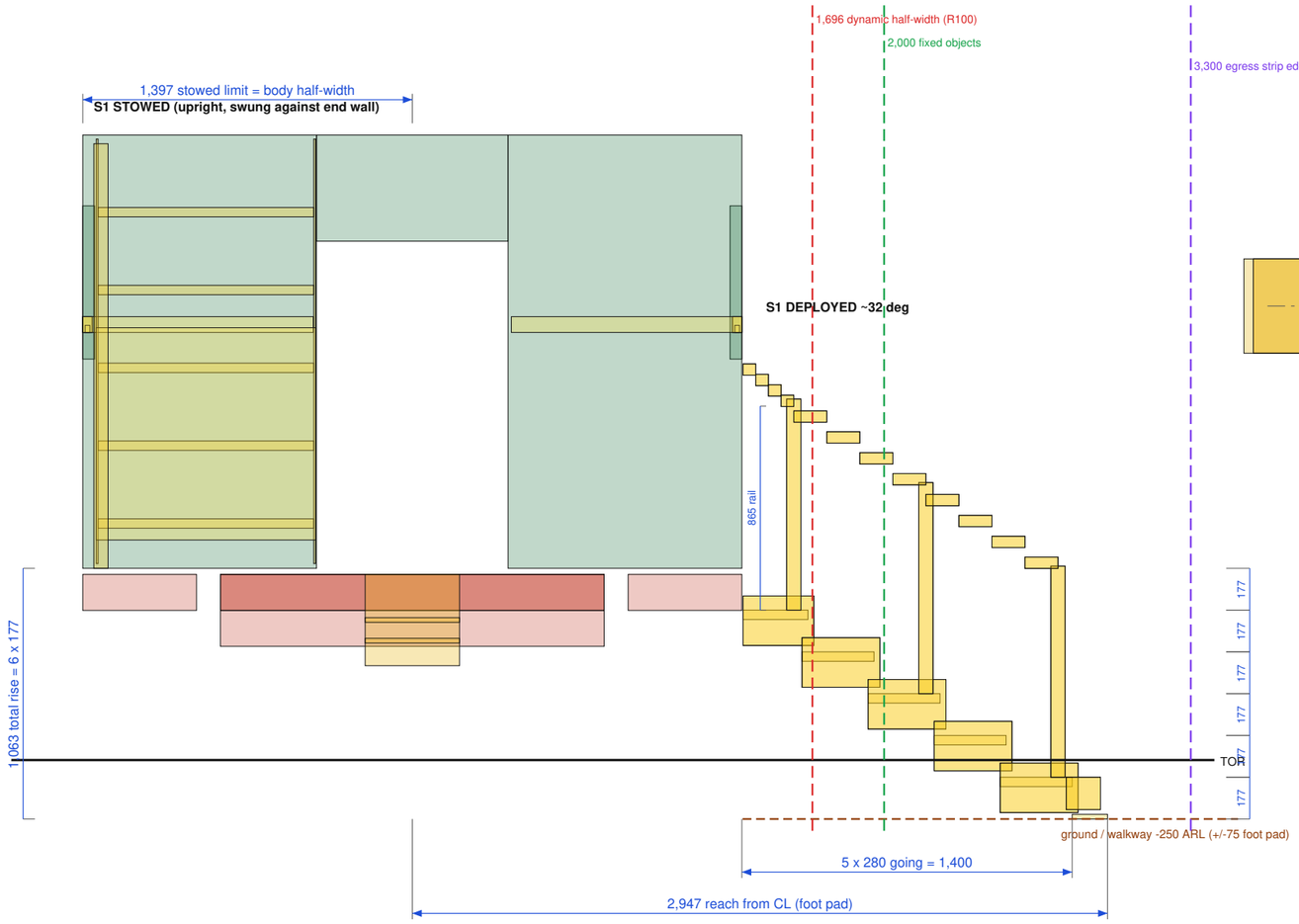
Sheet C-303 · REV L2 · 2026-10-27 · 1:20 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

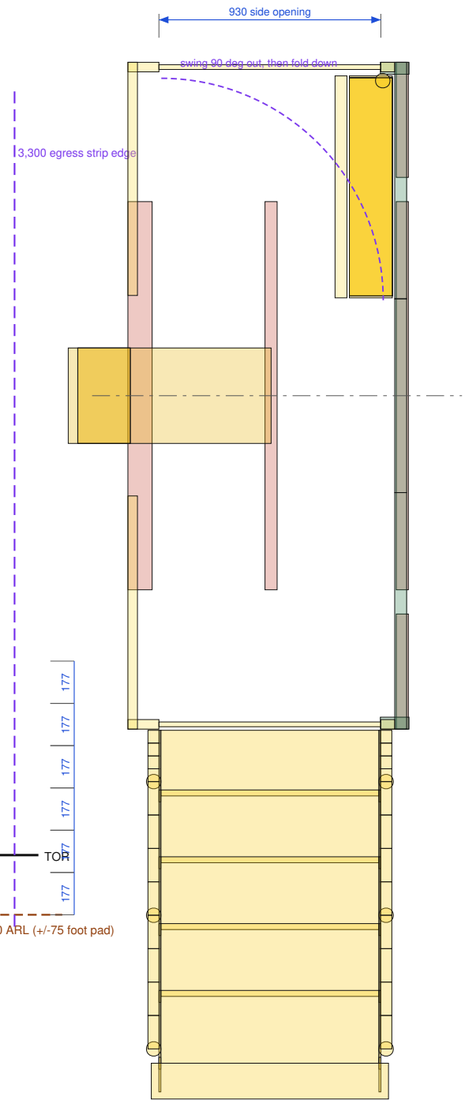
Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

SECTION AT END PLATFORM (looking forward): +Y stair DEPLOYED, -Y stair STOWED



PLAN - FRONT END PLATFORM



STAIR SCHEDULE

Item	Design (R2)
Count	18: S1 x16 (end platforms), S2 x2 (accessible side doors)
Risers	6 equal x 177 (7.0 in), floor 813 to ground -250 ARL
Treads	5 x 280 going (11 in), 3/16 in formed channel, abrasive nosing
Pitch / width	~32 deg / 914 (36 in) clear between stringers
Stringers	PL8 x 150 A36, hinge pins 2 x dia 25 4140 at floor edge
Handrails	both sides, 1-1/2 sch 40 at 865 on fold-flat 2 in sch 40 posts
Foot	adjustable pad +/-75; 2 x 3/8 G70 chain stays (foot unsupported)
Stowed	upright, inside 1,397 half-width: outline 1,696 UNCHANGED
Latch / interlock	gravity latch + safety pin + 'stowed' prox. switch in series
	loop: open = traction off + brake applied + cab lamp
Mass	~120 kg est.; gas-spring / damper assist AWAIT supplier
Finish	hot-dip galvanized, yellow nosings, 'EMERGENCY EXIT' decal

NOTES

Loads: treads/stringers 300 lb point + 100 psf; handrails 200 lb any direction (IBC 1607 - ASSUMED, PE to confirm basis incl. IBC 1011 / ASTM F770 / OSHA 1910.25). Calcs C15a-i all OK (max ratio 0.60, rail post).
S1: stowed upright against end wall beside 32 in end door (platform side opening stays clear for level boarding); crew/passenger pulls red release, swings stair into the opening, lowers it. S2: stowed beside 42 in door inside, slides across.
Egress: keep ground 2,000-3,300 from CL walkable on >= 1 side; wheelchair users - crew-assisted evacuation plan (County/PE).
Replaces the R1 fixed 2-tread side steps (265 / 275 risers, non-uniform, stopped 275 above rail).

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Coach emergency / boarding stairs (R2)

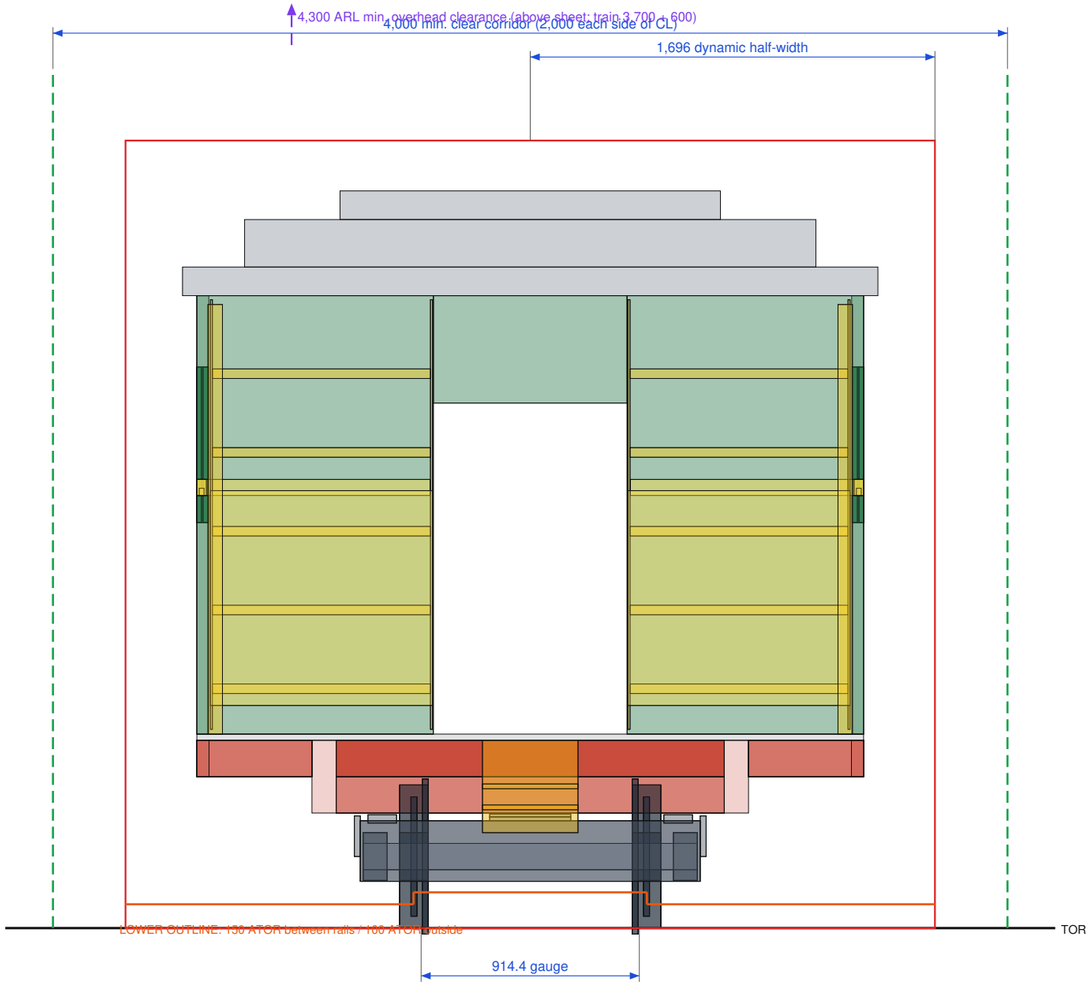
Sheet C-304 · REV L2 · 2026-10-27 · 1:25 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

COACH IN CLEARANCE OUTLINE (end view)



CLEARANCE BUILD-UP

Item	mm
Static half-width (coach + eave)	1,457
Coach end throw, R100	91
Coach centre throw, R100 (inside)	38
Loco end throw, R100	148
Body sway 2° at roof	90
Dynamic half-width	1,696
Adopted fixed-object clearance	2,000 from CL

TRACK NOTES

Rail: 60 lb bolted relay rail (RULED) - DO NOT ORDER until spec approved + PE
approves rail, joint bars, bolts and drilling (I-17, I-19).
Gauge 914.4 (36 in) +6/-3; widen 0 at R100.
Min. radius 100 m (Alan Keef, 23 Oct): stations on tangent track.
Design grade 2% GOVERNING (max 2.0% tangents; curves compensated so grade + curve resistance <= 2.0%).
Max wheel load basis 26 kN (L4: max driver wheel 25.89 kN).
Platforms OPTIONAL (R2): if built, 813 ARL, edge 1,472 from CL, tangent.
Egress strip: ground 2,000-3,300 from CL clear + walkable (+/-75)
on >= 1 side for coach stairs (reach 2,947); else sign 'do not deploy'.
Overhead: min. 4,300 ARL within 2,000 of CL (train 3,700 + 600).
Ties, fastenings, ballast section: AWAIT PE / supplier.
Loop sketch 5,180 ft vs 6,000 ft budget (survey pending).
LOWER OUTLINE: no part (wheels excepted) below 150 ATOR between the
rails or 100 ATOR outside; basis level track + vertical curves
R >= 500 m, new wheels, static. Min. found 196 mm (K2-lo).
FOULING POINT: clearance mark at 3.6 m track centres (2 x 1,696 + 200),
34.0 m from the points of each #8 (checked 33.99 m, K2-fp).
Working layout F (R80, level) - OWNER PENDING.

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Track, rail and clearance outline

Sheet K-401 · REV L2 · 2026-10-27 · NTS / 1:25 end

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

LOCOMOTIVE BOM (parts grouped by CAD name; masses = model volume × 7,850 kg/m³, steel only)

Item	Description (from CAD model PF-L100)	Qty	Material	kg (model)
1	main rail HSS12x6x3/8 (LT18: Y±240, Z1150-1454.8)	2	A500 Gr C	1,389
2	front end sill HSS12x6x3/8	1	A500 Gr C	21
3	rear end sill HSS12x6x3/8	1	A500 Gr C	21
4	crossmember HSS8x4x1/4	5	A500 Gr C	48
5	body bolster box PL12/PL10	2	A36	78
6	body centre plate Ø300 PL40 (truck top Z1110, LT18)	2	A36	44
7	running-board outrigger C3x4.1	14	A36	554
8	running-board edge angle L3x3x1/4	2	A36	230
9	running board / deck PL6 tread plate (top Z1,605, LT11)	1	A36	1,248
10	loco rear drop drawbar box PL16	1	A572 Gr 50	155
11	loco rear pocket top plate PL20	1	A572 Gr 50	14
12	loco rear pocket bottom plate PL20	1	A572 Gr 50	14
13	pilot / buffer beam PL12 box	1	A36	-
14	ballast pocket PL12 + shear lugs (750 kg max each side)	2	A572 Gr 50	128
15	ballast plates 800x480x25 A36, up to 9 per side (removable)	2	A36	1,441
16	axle Ø1450 driver	4	EA4T / 4140	467
17	wheel tread rim 50 deep Ø1450 driver	8	EN 13262 ER7 / AAR B	1,657
18	wheel web 25 Ø1450 driver	8	EN 13262 ER7 / AAR B	2,198
19	wheel hub OD200 Ø1450 driver	8	EN 13262 ER7 / AAR B	137
20	flange Ø1450 driver	8	EN 13262 ER7	182
21	shrunk sprocket hub 4140 Ø240x110 + Ø350 flange on raised Ø150 axle seat, SHRI…	4	AISI 4140 Q&T	95
22	hub flange Ø350 x 20 integral (L4.1), 12 x Ø17 reamed on PCD 300	4	EN 13262 ER7	32
23	45T ANSI 120-2 split rim sprocket ring with web PD546 OD569 (Source Bot rev 3)…	4	purchased	-
24	oil-bath chain case 3 mm + floating seal plate at the axle (ISO VG 220, ~4 L)	4	A36 3 mm + seal plate (L-103.2)	-
25	15T ANSI 120-2 motor sprocket Martin D120B15 PD183 OD202, hub broached DIN 548…	4	purchased	-
26	Poclain MS05 468/234 cc 2-speed HighFlow + T04 brake, shaft code 5 DIN 5480 W5…	4	purchased	-
27	motor bracket PL25 slotted (4 x M20 in 50 mm slots, 2 jack screws) welded to t…	4	A572 Gr 50 PL25 (L-103.2)	127
28	motor bracket gusset/top plate PL20 to transom	4	A572 Gr 50 PL20 (L-103.2)	5
29	brake disc Ø340x25 on the axle (Y+, unchanged)	4	purchased	-
30	spring-applied air-released caliper (2 per disc, ref. Coremo D 3.5N)	8	purchased	-
31	inside axlebox + primary guide (AWAIT bearing)	8	purchased	-
32	primary spring Ø140 reservation (LT35 AWAIT)	8	purchased	-
33	crank pin boss (decorative)	8	A36 (decorative)	2
34	inside sideframe top chord PL30 (top Z1110, LT18)	4	A572 Gr 50	228
35	inside sideframe bottom chord PL30	4	A572 Gr 50	166
36	horn cheek PL30	16	A572 Gr 50	48
37	side rod (bronze look, decorative, static phase)	4	A36 (decorative)	107
38	truck transom box PL16	2	A36 / A500 Gr C	155
39	truck bolster / pivot seat PL16	2	A36	17
40	traction pin Ø120 4140 Q&T, Z650-1110 (L4.2)	2	AISI 4140 Q&T	82
41	traction-pin bearing housing GE120ES (AWAIT SKF rating) at Z650 under transom …	2	purchased	-
42	front-truck cylinder-support extension PL30	2	A572 Gr 50	45
43	cylinder support arm PL25 (lengthened for Y±790)	2	A36	30
44	decorative cylinder Ø380 at Y±790 (moved from ±750:-wheel clash K2-h, route ch…	2	A1011 CS (decorative)	-
45	axle	2	EA4T / 4140	191
46	wheel tread rim 50 deep	4	EN 13262 ER7 / AAR B	355
47	wheel web 25	4	EN 13262 ER7 / AAR B	162
48	wheel hub OD200	4	EN 13262 ER7 / AAR B	83
49	flange	4	EN 13262 ER7	42
50	leading axlebox (inside)	4	purchased	-
51	leading-truck inside sideframe PL25	2	A36	134
52	leading-truck transom	1	A36 / A500 Gr C	73
53	pony ballast 0.625 t per side (steel plates on the sprung pony frame, own seco…	2	purchased	-
54	leading-truck swing arm 1,075 (LT18)	1	A36	49
55	swing-arm anchor hanger PL20 box	1	A36	-
56	boiler-style hood shell 3 mm (cosmetic, removable)	1	A1011 CS 3 mm	-
57	smokebox front	1	A1011 CS	-
58	stack (cosmetic)	1	A1011 CS	-
59	dome (cosmetic)	1	A1011 CS	-
60	cab lower walls 3 mm on HSS2x2x1/8	1	A500 Gr C	-
61	cab front post HSS4x4x1/4	2	A500 Gr C	215
62	cab rear post HSS4x4x1/4	2	A500 Gr C	215
63	cab front wall / windows	1	PC glazing on A500 frame	-
64	cab roof 3 mm skin on HSS3x2x1/8 purlins @600 (3 in deep, adopted R2)	1	A1011 CS skin on HSS3x2x1/8 purlins	
65	cab step tread PL8 grip strut	8	A36	60
66	step stringer FB 20x75	8	A36	113
67	grab iron Ø32 solid A36	4	A36	30

Purchased items: mass per supplier (AWAIT). '-' = shell / wall / roof / envelope modelled solid in CAD - mass from the ledger, not the model. L4 drive units (Poclain MS05 + T04, RS120-2 chain 15T:45T, pressed hub, oil-bath case, disc/2 calipers) detailed on L-103.2 (×4; shop detail Loco Bot L-103). Front ballast 150.8 kg (2 × 75.4 kg plates, 1 per side) + BT-1 battery 68.2 kg in the Y… ell leg
CAD steel total PF-L100: see cad/mass_report.json. Loco ledger rev L4.2 incl. barrel 2,280.6, traction pins Ø120 (+22 kg each vs Ø100), NB-1 59.8 kg, final drive 336.7 kg/axle, ballast and pony ballast: SUPERSEDED - current 20,404.2 kg pending freeze (data/mass_ledger.json).

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Bill of materials - locomotive PF-L100 (rev L4.2)

Sheet S-501 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.

TENDER T2 (PF-T200)

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	tender main rail HSS10x4x1/4	2	A500 Gr C	301
2	tender crossmember HSS10x4x1/4	3	A500 Gr C	133
3	tender fish-belly bolster top flange PL12	2	A36	73
4	tender fish-belly bolster web PL10 (deep at centre plate)	4	A36	110
5	tender fish-belly bolster web PL10 (shallow outboard)	8	A36	53
6	tender fish-belly bolster bottom flange PL12 (centre-plate seat)	2	A36	34
7	axle	4	EA4T / 4140	418
8	wheel tread rim 50 deep	8	EN 13262 ER7 / AAR B	651
9	wheel web 25	8	EN 13262 ER7 / AAR B	259
10	wheel hub OD200	8	EN 13262 ER7 / AAR B	165
11	flange	8	EN 13262 ER7	77
12	axlebox	8	purchased	-
13	sideframe PL25	4	A36	240
14	transom HSS10x4x1/4 (upright, pair)	4	A36 / A500 Gr C	195
15	transom top cap PL16	2	A36	29
16	centre plate Ø340 PL12 + Ø60 centre pin	2	A36	17
17	side bearer	4	A36 + UHMW pad	16
18	tender front drop drawbar box PL16	1	A572 Gr 50	142
19	tender front pocket top plate PL20	1	A572 Gr 50	14
20	tender front pocket bottom plate PL20	1	A572 Gr 50	14
21	tender rear drop drawbar box PL16	1	A572 Gr 50	142
22	tender rear pocket top plate PL20	1	A572 Gr 50	14
23	tender rear pocket bottom plate PL20	1	A572 Gr 50	14
24	tender deck 6 mm tread plate (sealed metal spill floor)	1	A36 tread plate	539
25	tender deck cross-bearer C6x8.2 (outboard) web	14	A36	47
26	tender deck cross-bearer C6x8.2 (outboard) top flange	14	A36	23
27	tender deck cross-bearer C6x8.2 (outboard) bottom flange	14	A36	23
28	tender deck cross-bearer C6x8.2 (between rails) web	4	A36	32
29	tender deck cross-bearer C6x8.2 (between rails) top flange	4	A36	16
30	tender deck cross-bearer C6x8.2 (between rails) bottom flange	4	A36	16
31	tender side skin 10 ga (3.4 mm) on posts @500 + 50 mm lining (riveted-plate lo…	2	A1011 CS	-
32	tender end skin 10 ga (3.4 mm) on posts @500 + 50 mm lining	2	A1011 CS	-
33	tender flat roof 10 ga (3.4 mm) on HSS2x2x1/8 bows @500 (sealed)	1	A1011 CS on HSS bows	-
34	tender side post HSS2x2x1/8 @500	20	A500 Gr C	115
35	tender roof bow HSS2x2x1/8 @500	10	A500 Gr C	-
36	tender end post HSS2x2x1/8 @500	8	A500 Gr C	46
37	coal-effect cover top 1.5 mm (16 ga) formed steel, removable (height ASSUMED)	1	A1011 CS 16 ga	-
38	coal-effect cover side skirt 1.5 mm	2	A1011 CS 16 ga	-
39	coal-effect cover end skirt 1.5 mm	2	A1011 CS 16 ga	-
40	tender ENGINE: AIE Kubota V3307-TIE4AB-KEA-1 + radiator + H1P053 pump (rev L4 …	1	purchased	-
41	tender GENERATOR: genset PowerTech PTI-38 open-frame 1708x667x864, 699 kg in r…	1	purchased	-
42	tender GEN-MOUNT: generator mount allowance 80 mm	1	purchased	-
43	tender COMPRESSOR: IR 2340 14 cfm @175 psi, 68 kg, ~3.7 kW, electric clutch, b…	1	purchased	-
44	tender FUEL: custom 150 L 5052 Al diesel tank X3900-4500 in its own compartmen…	1	purchased	-
45	tender FIREWALL: steel firewall plate at X3800, full height, 3 mm A36 (fire se…	1	A36	-
46	tender START-BATT: genset start battery ODX-AGM34 in vented box + master switc…	1	purchased	-
47	tender MAIN-RES: main reservoirs 3 x Midland Haldex 19820 (47 L, 150 psi) long…	1	purchased	-
48	tender RESERVOIR: hydraulic reservoir 600x600x450	1	purchased	-
49	tender COOLER: hydraulic oil cooler >= 14 kW @ 43 C outside air (rev L4), LEFT…	1	purchased	-
50	tender CABINET: RIGHT-wall outward-facing NEMA 3R electrical cabinet: panel QO…	1	purchased	-
51	tender PARTITION: tank-bay partition 50 mm at x2800	1	A36 + insulation	-
52	tender XFMR: shore isolation transformer Acme T2535173S 15 kVA UL 506 NEMA 3R …	1	purchased	-
53	tender XFMR-ISO: 4 elastomer isolators + captive bolts + catcher strap, 3.5 kg…	1	see note	35
54	tender XFMR-CROSS-F: cradle cross angle L3x3x1/4 fwd, rail inner face to inner…	1	A36	59
55	tender XFMR-CROSS-R: cradle cross angle L3x3x1/4 aft, top Z380	1	A36	59
56	tender XFMR-BOXRAIL-1: inrush box rail L3x3x1/4 (288) between the cross angles…	1	A36	13
57	tender XFMR-BOXRAIL-2: inrush box rail L3x3x1/4 (288), right side (TB-3.4)	1	A36	13
58	tender XFMR-HANGER-FL: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
59	tender XFMR-TAB-FL: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
60	tender XFMR-HANGER-FR: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
61	tender XFMR-TAB-FR: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
62	tender XFMR-HANGER-RL: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
63	tender XFMR-TAB-RL: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
64	tender XFMR-HANGER-RR: cradle hanger L3x3x1/4 (~709), PL10 150x100 tab welded …	1	A500 Gr C	-
65	tender XFMR-TAB-RR: hanger tab PL10 150x100 (bolts 2 x M16 at Z963) + outboard…	1	A36	1
66	tender XFMR-SHIELD: bolted 3 mm steel ballast/spray shield 450 x 1,120 (widene…	1	EN 13262 ER7	-
67	tender INRUSH: inrush limiter (BDESB/MP-ESB-50 class, 120 V coil) in NEMA 4X ~…	1	see note	148

COACH R2 regular (PF-C300) - per coach; accessible coach PF-C300A per CAD

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	coach main rail HSS12x4x1/4	2	A500 Gr C	798
2	end sill HSS12x4x1/4	2	A500 Gr C	128
3	body bolster HSS12x4x1/4 + PL10 doublers	2	A500 Gr C	128
4	axle	4	EA4T / 4140	418
5	wheel tread rim 50 deep	8	EN 13262 ER7 / AAR B	651
6	wheel web 25	8	EN 13262 ER7 / AAR B	259
7	wheel hub OD200	8	EN 13262 ER7 / AAR B	165
8	flange	8	EN 13262 ER7	77
9	axlebox	8	purchased	-
10	sideframe PL25	4	A36	240
11	transom HSS10x4x1/4 (upright, pair)	4	A36 / A500 Gr C	195
12	transom top cap PL16	2	A36	29
13	centre plate Ø340 PL12 + Ø60 centre pin	2	A36	17
14	side bearer	4	A36 + UHMW pad	16
15	crossmember HSS6x2x3/16	10	A500 Gr C	236
16	outrigger HSS6x2x3/16	10	A500 Gr C	70
17	side sill HSS6x2x3/16	2	A500 Gr C	295
18	coach front drop drawbar box PL16	1	A572 Gr 50	161
19	coach front pocket top plate PL20	1	A572 Gr 50	14
20	coach front pocket bottom plate PL20	1	A572 Gr 50	14
21	coach rear drop drawbar box PL16	1	A572 Gr 50	161
22	coach rear pocket top plate PL20	1	A572 Gr 50	14
23	coach rear pocket bottom plate PL20	1	A572 Gr 50	14
24	floor 1 in composite panel	1	purchased	-
25	side wall lower (HSS2x2x1/8 frame + skin)	2	A1011 CS on HSS2x2x1/8	-
26	side wall upper / letter board	2	A1011 CS	-
27	window post HSS2x2x1/8	18	A500 Gr C	-
28	window (polycarbonate)	16	purchased	-
29	end wall (32 in centred door)	4	A1011 CS on HSS	-
30	end wall over door	2	A1011 CS on HSS	-
31	roof eave	1	A1011 CS on HSS bows	-
32	roof crown	1	A1011 CS on HSS bows	-
33	roof clerestory	1	A1011 CS on HSS bows	-
34	platform guard top rail 42 in	8	A53 B sch 40 (2 in, C10d)	16
35	side opening gate bar 42 in (latched, interlocked)	4	A53 B sch 40	18
36	end gate rail	4	A53 B sch 40	82
37	S1 stair stowed - stringer	8	A36 / A500 Gr C (C-304)	-
38	S1 stair stowed - tread	20	A36 / A500 Gr C (C-304)	-
39	S1 stowed handrails (folded flat)	4	A53 B	-
40	S1 vertical swing pivot post 2-1/2 in sch 80 + stowed latch / proximity switch	4	A53 B	76

DRAWBARS / PINS (train set)

Item	Description (CAD PF-T200 / PF-C300 / PF-K400)	Qty	Material	kg (model)
1	DB-800 loco-tender drawbar PL32x150 A572-50	1	A572 Gr 50	35
2	DB-800 loco-tender pin Ø60 4140	2	AISI 4140 Q&T	11
3	DB-1473 #1 drawbar PL32x150 A572-50	1	A572 Gr 50	60
4	DB-1473 #1 pin Ø60 4140	2	AISI 4140 Q&T	11
5	DB-1473 #2 drawbar PL32x150 A572-50	1	A572 Gr 50	60
6	DB-1473 #2 pin Ø60 4140	2	AISI 4140 Q&T	11
7	DB-1473 #3 drawbar PL32x150 A572-50	1	A572 Gr 50	60
8	DB-1473 #3 pin Ø60 4140	2	AISI 4140 Q&T	11
9	DB-1473 #4 drawbar PL32x150 A572-50	1	A572 Gr 50	60
10	DB-1473 #4 pin Ø60 4140	2	AISI 4140 Q&T	11

PE STAMP / SIGNATURE

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'-' in kg = purchased, or shell/wall/roof envelope modelled solid in CAD (mass from ledger). Coach R2 adds (not separately modelled): roof bows HSS3x2x1/8 @880 (R1-f), guard posts 2 in sch 40 (C10d), ERV HE05IN roof pod + heat pump per coach (E-701).
Drawgear purchase set (5 joints): 12 pins Ø60 × 150 under head 4140 Q&T (10 + 2 spares) with bushes, Ø10 cross-bolts, R-clips; 10 safety chains 5/8 in Grade 100 (2 per joint) + 20 Crosby G-2130 1 in shackles (+2 spares).

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

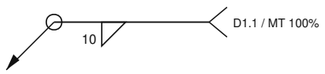
Bill of materials - tender T2, coach R2, drawgear

Sheet S-502 · REV L4 · 2026-10-28 · NTS

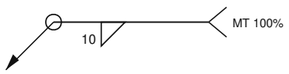
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

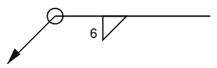
TYPICAL WELD SYMBOLS (AWS A2.4) - arrow side below reference line



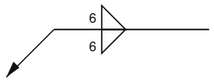
Drop drawbar box to rails/sill



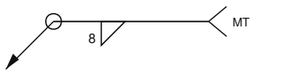
Pocket plates PL20 to box



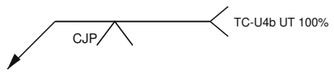
Crossmembers to main rails



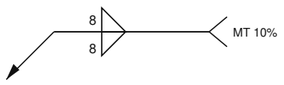
Bolster box seams (both sides)



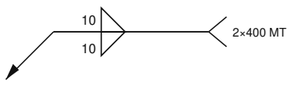
Centre plate to bolster



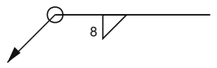
Horn cheek to chord (powered truck)



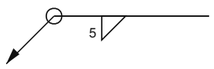
Transom box seams



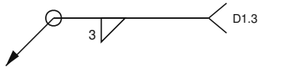
Motor bracket PL25 to transom



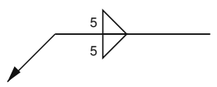
Engine / genset pedestals to tender rails



Coach crossmembers / outriggers



Window posts, roof bows HSS3x2x1/8



Stair stringers / treads (C-304)

Fillet sizes = leg (mm). ○ = weld all around. Tail = inspection. E70XX, prequalified WPS (D1.1 Cl. 5); preheat per Table 5.8 for A572-50 > 20 mm; grind toes on cyclic truck welds (cat. C→B').

L4.2: safety-chain lugs PL44.5 (1-3/4 in) A572-50 faced to 42.5 mm at the jaw, 1-1/8 in Crosby G-2130 shackles + S-1325A; CJP (TC-U4b) + PL16 diaphragm; UT 100 %; preheat Table 5.8.

NOTES

1. Running-gear and drive fasteners: re-torque after first 10 h running; witness-mark all.
2. Structural slip-critical joints (if any added): ASTM F3125 A325, RCSC 2020 turn-of-nut.
3. CWI inspection of primary members (drawgear, trucks, drive brackets). Welders qualified to D1.1 / D1.3.
4. Zinc-plated fasteners: use dry K 0.20 column; lubricated (MoS2) K 0.14 column only where specified.
5. L4.2 P-103-1 motor hanger: L-weld 140 + 150, 10 mm fillet both sides + 2 PL16 gussets (6 mm both sides, structural); MT 100 % (Loco Bot L-103 fatigue check).

FASTENER TORQUE TABLE (ISO 898-1, preload 0.75 × proof)

Size	Class	As mm²	Preload kN	T dry K0.20 N·m	T lub K0.14 N·m
M10	8.8	58.0	25.2	50	35
M10	10.9	58.0	36.1	72	51
M12	8.8	84.3	36.7	88	62
M12	10.9	84.3	52.5	126	88
M16	8.8	157.0	68.3	219	153
M16	10.9	157.0	97.7	313	219
M20	8.8	245.0	110.2	441	309
M20	10.9	245.0	152.5	610	427
M24	8.8	353.0	158.8	762	534
M24	10.9	353.0	219.7	1055	738

APPLICATION SCHEDULE

Joint	Bolt	N·m dry K0.20	Locking
Axlebox keepers	M20 10.9	610	Nord-Lock
Truck pivot cap	M24 10.9	1,055	witness mark
Motor slide bolts	M20 10.9	610	Nord-Lock; after jack-screw set
Brake caliper mounts	M20 10.9	610	Loctite 243
45T split ring to hub flange (fitted DIN 609)	M16 8.8	219	Nord-Lock; re-torque 10 h
MS05 mounting flange	10.9	per Poclain	AWAIT
Engine / genset isolators	M16 8.8	219	prevailing-torque nut
Ballast pocket rods (P-106)	M20 8.8	120	Nord-Lock
Clamp-bar set screws 106-CB1	M16	4.8	jam nut 60 + Loctite 243
Seat anchors	M12 8.8	88	through-bolt + backing PL
Stair pivot posts	M16 8.8	219	Nord-Lock
Axle end caps	M16 10.9	313	tab washer
Cab to frame	M16 8.8	219	Nord-Lock

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Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Weld and fastener schedule (AWS A2.4 symbols, torques)

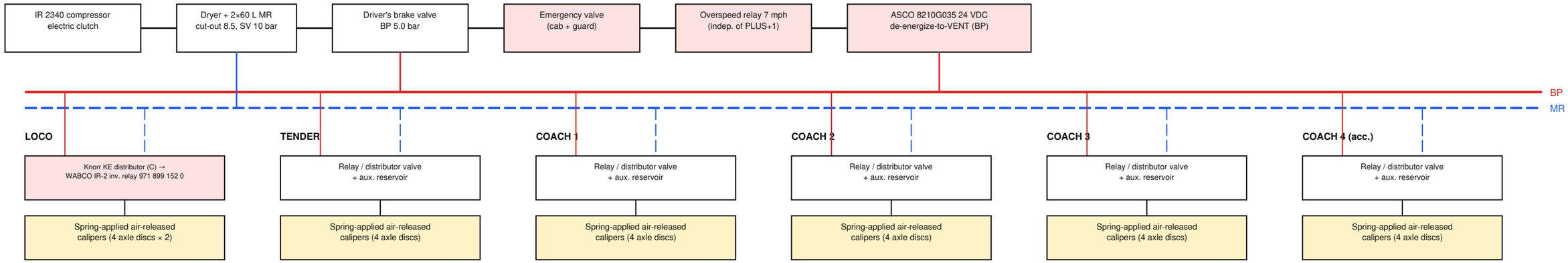
Sheet S-503 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
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Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

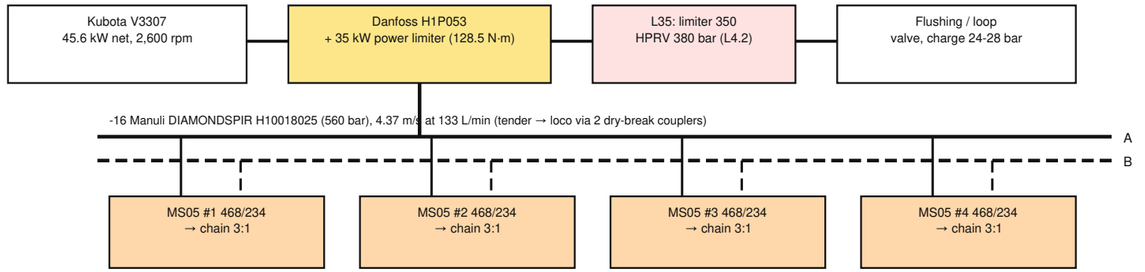
AIR BRAKE (automatic, BP 5.0 bar / 72.5 psi UIC; fail-safe: loss of BP, MR or 24 V applies spring brakes on every axle except the unbraked pony)



BRAKE PERFORMANCE (calc §5)

Case (loaded train, 2% down)	Result	Criterion
Emergency force (tare-based 12% ratio)	67.5 kN	< wet adhesion of braked tare 80.2 kN
Emergency decel. level / 2% down	0.828 / 0.632 m/s²	-
Emergency stop from 6 mph (governed)	9.7 m	≤ 20 m (ASSUMED)
Emergency stop from 7 mph (overspeed trip)	12.4 m	≤ 30 m
Service stop (70%) from 6 mph	13.4 m	≤ 30 m
Parking: all spring brakes, FS 1.5 (T04 = parking/holding only, 0 in motion)	24.0 kN req. vs 64.1	OK - loco alone NG → rule + chocks
Loco axle-disc clamp per caliper (2/disc)	19.8 kN	spec ≥ 22 kN
In-train buff, loco-tender	4.1 kN	≤ 70 kN

HYDRAULIC TRANSMISSION (closed loop, ISO 4413)



FAIL-SAFE LOGIC (loco, rev L4 - PE to confirm)

BP drop (driver valve, emergency valve, train parting) → KE raises C → IR-2 EXHAUSTS the spring chambers → loco springs apply.
Loss of 24 V, or Pilz PNOZ s30 7 mph trip → ASCO 8210G035 de-energizes → BP VENTS → emergency on every vehicle; the same s30 safe output drops the H1P053 CCO (control cut-off) → pump to neutral. Loss of MR / IR-2 supply → chambers exhaust → APPLY.
Release needs BP ≥ 4.8 bar AND MR AND 24 V healthy. IR-2 supply regulated ≤ 6 bar (Coremo release max) - verify IR-2 data sheet.

DRIVE NOTES

Traction (L4): 2% start need 32.61 kN; 37.67 kN at 350 bar; NO flow divider - PLUS+1 traction control (pony ref., destroke); NO SAND (deleted). Reverse pull cap 28.9 kN; reverse full-train starts ≤ 1.20 % (L4.2). Sanders DELETED.
GOVERNED 6 mph: H1P053 stroke stop 42 cc sealed + HARD tamper-sealed 2,600 rpm engine limit (hardware 7.64 mph). 7 mph trip: 2-channel speed relay independent of PLUS+1 → BP vent + pump neutral.
ANY TRIP / 24 V loss (L4.2): swash ramp ≥ 3 s + loop BYPASS normally open = FAIL-TO-FLOAT (1st-gear float sized); 2-ch T04 switch vents BP first;
T04 choke Ø2.4 + 1 L acc. + Ø1.2 orifice per port, hard/4SH lines downstream (run-in ≤ 33.2 kN). Direction change only at zero speed.
T04 RELEASE (L4.2): H1 CHARGE port M3 → check + 10 µm → CDK 20 bar reducing/relieving → NV 3/2 → choke → 4 × T04 (loop shuttle DELETED). Release ≤ 8 s (met).
Manual release: check → 6.0 L bladder acc. (13.5 bar) + 12 cm³ hand pump → guarded padlocked 3/2 (limit sw. → traction inhibit); T04 M12 screws last resort.
35 kW anti-stall (34.5 kW while the compressor clutch is in; clutch only at low demand or MR < 7.0 bar). H1P053 with CCO (to the s30 trip); external charge filter HYDAC DF 60.
TOWING: tow valve (KHB3K DN06, 500 bar) open, T04 release by 6 L acc. + hand pump, max 1 mph (E11-S5 ruling 6). DESCENTS in FLOAT (bypass open, ≥ 3 s ramp) on friction brakes sized for 2 %. Draft gear: preloaded, k 20-100 kN/mm, renew at 1.0 mm TOTAL play. Loco brake ≤ 0.69 m/s².
LOCO RELEASE ≤ 4 s (L4.2: 2.22 / 2.36 s): 8 × 1.0 L calipers; C_spec ≥ 3.2 per branch; QEV option B (maker-cert. ≥ 15 bar): relieving regulator 1/2 in C ≥ 10 @ 6.0 bar → 1/2 in relay C ≥ 8 (piloted by IR-2) →
3/4 in trunk ID ≥ 19 C ≥ 8 → 4 × 1/2 in hose ≤ 2 m C ≥ 2.0 to both calipers; quick-exhaust per axle; release-proof 5.1 bar + coach release 2-ch (E3.6h) → traction enable. Test ≤ 4.0 s at MR cut-in.
Loop hose Manuli DIAMONDSPIR H10018025 (560 bar WP, ISO 18752-C, OD 41.2, min bend 350 ≤ tender R70 394.3). Loop parts ≥ 420 bar WP ≥ HPRV 380;
spike ~430 covered by ISO 18752 / ISO 6605 impulse rating; strength checked at 430; whip restraints; system test per ISO 4413 (PE), not above 420.

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
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Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Brake, air and hydraulic schematic

Sheet B-601 · REV L4.2 · 2026-10-28 · NTS

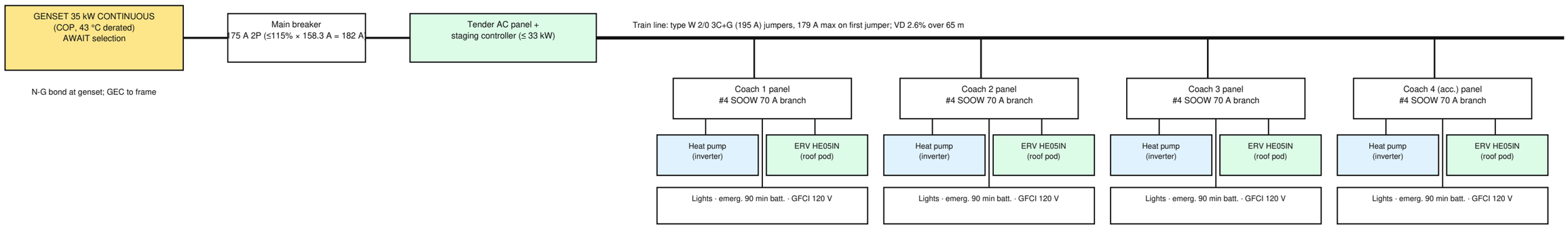
Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
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PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Speed: 6 mph governed limit · 6.5 mph HMI warning · 7 mph independent trip · 3 mph max if cameras fail. Not PE-reviewed.

ONE-LINE (120/240 V 1-ph 60 Hz, separately derived, NFPA 70-2023 Art. 445/250/400/522/525)



LOAD AND HVAC BALANCE (calc §8, §10)

Item	Value	Note
Hot-humid demand (plan coach-climate)	39.7 kW	plan basis; 26.8 / 31.7 kW other cases
Outside air, 4 coaches (62.1)	454.0 L/s	279 / 126 cfm per coach; plan 848 L/s kept for G1-b
OA sensible / latent load	10.7 / 8.2 kW	0.4% design day
ERV saving (ε 0.55 ASSUMED)	8.9 kW electric	COP 2.5; ε needed 0.344
Demand WITH ERVs	31.1 / 27.7 kW	plan OA / 62.1 OA; ≤ 34.4 kW target
Staging controller	≤ 33 kW	sheds compressor stages; keeps vent + emergency lights
Rev L4 running load hot / winter	30.94-32.44 / 31.20-32.70 kW	under the 33 kW shed point (Electro E3)
Genset current 35 kW / 240 V	145.8 A	main breaker 175 A
Alternative	45 kW set	needs tender bay re-layout (LT45.1)

NOTES

1. Equipment grounding conductor continuous through every jumper; bonding jumpers across truck pivots.
2. Jumpers: inline locking connectors rated ≥ 200 A, strain relief, interlocked so no live pins exposed.
3. Genset 35 kW continuous (ISO 8528 COP) 1-ph, ≤ 900 kg; tender reservation X1060-2768 (≤ 1,775) × 800 × 1,000 (rev TB-3, basis PTI-38). Cabinet: tender R wall NEMA 3R X2820-3780; listed 200 A 3R UL 1008 ATS w/ integral main; main bkr QOM22225NRB encl. handle ≤ Z1639.
4. ERV RenewAire HE05IN 120 V (880×956×307, 43 kg) in low roof pod ≤ 3,450 ARL; OA/EA louvres separated; MERV 13.
5. Emergency lighting and exit marking battery-backed 90 min per coach (plan EL1).
6. Electrical design to be reviewed/inspected per NFPA 70 and County; wiring by licensed electrician.

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Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Electrical one-line and coach HVAC (35 kW + ERV)

Sheet E-701 · REV L4 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

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PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Not PE-reviewed.